# SIMULTANEOUS CONFIDENCE BANDS FOR FUNCTIONAL VARYING-COEFFICIENT MODELS

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*Abstract:* We develop quantitative simultaneous inference for functional varying-coefficient models under sparse and dense sampling. A subject-level linear representation connects local polynomial estimation to high-dimensional Gaussian approximation without reconstructing individual trajectories. Explicit error bounds account for dimension, sampling frequency, smoothing bias, estimated designs, and fitted residuals. A Gaussian multiplier bootstrap calibrates joint confidence bands over time and coefficients. The theory allows growing regression dimension and extends to high-dimensional functional means. Simulations and a liver-disease biomarker analysis illustrate the procedure.

*Key words and phrases:* Functional data, Gaussian approximation, multiplier bootstrap, simultaneous inference, varying-coefficient model.

## 1. Introduction

Simultaneous inference for functional data requires uncertainty to be controlled over an entire time interval and, for multivariate targets, across several functions at once. This problem is particularly challenging under sparse and irregular sampling, when individual trajectories cannot be accurately reconstructed from their own measurements. Varying-coefficient models (VCMs) provide an important setting: they describe associations that evolve over time through a vector of coefficient functions (Hastie and Tibshirani, 1993; Gu et al., 2014; Cao, Liu and Zhou, 2018). Joint confidence bands must account for dependence within subjects and across coefficient estimates, while remaining valid as the number of observations per subject increases. We study this problem through quantitative Gaussian and bootstrap approximation, with explicit control of the errors that determine simultaneous coverage.

The sampling design has long been central to functional data analysis; see Wang, Chiou and Müller (2016). Dense-data inference can exploit accurate trajectory reconstruction (Zhang and Chen, 2007), whereas sparse-data methods pool information across subjects (Yao, Müller and Wang, 2005). Uniform estimation theory and optimal mean-estimation rates clarify the transition between these settings (Li and Hsing, 2010; Cai and Yuan,

2011). Zhang and Wang (2016) also distinguish equal-subject and equal-observation weighting when counts differ. These choices change how sampling frequency enters the variance and cannot be interchanged without checking the resulting theory.

Simultaneous inference is available in both classical regimes. For mean curves, dense-data bands include Degras (2011); Cao, Yang and Todem (2012), and sparse-data bands include Ma, Yang and Carroll (2012); Zheng, Yang and Härdle (2014). For regression, local polynomial and spline methods provide established estimation and inference tools (Wu and Chiang, 2000; Huang, Wu and Zhou, 2004; Fan and Zhang, 2000). Gu et al. (2014) develop sparse-data coefficient bands with time-independent covariates, while Cao, Liu and Zhou (2018) allow time-varying covariates under sparse irregular sampling. Dense functional-response regression is studied by Zhu, Li and Kong (2012); Chang, Lin and Ogden (2017). Zhou et al. (2024) develop empirical-likelihood M-estimation and simultaneous regions for functional responses with subject-level covariates observed on a common grid. More recently, Chang and McKeague (2026) construct empirical-likelihood bands for concurrent regression on a dense fixed grid, allowing nonsmooth functional features. These dense-grid results and sparse random-design results address different observation schemes.

Several inferential methods accommodate a range of observation frequencies. Kim and Zhao (2013) study longitudinal mean inference, Chen and Yao (2017) develop self-normalized pointwise inference for varying coefficients, and Wang et al. (2018) provide empirical-likelihood pointwise regions and global tests for concurrent functional regression. These pointwise and global procedures address different inferential targets from bands calibrated jointly over time and coefficient functions.

For scalar mean functions, Cai and Hu (2025) and Lou and Wu (2026) develop unified simultaneous inference using B-spline and local linear estimators, respectively. Their Gaussian approximations yield asymptotically valid bands and characterize covariance transitions across sampling frequencies. Our focus is quantitative simultaneous inference for vector-valued functional parameters throughout this transition.

We take a nonasymptotic approach based on high-dimensional Gaussian approximation and comparison theory (Chernozhukov, Chetverikov and Kato, 2017; Chernozhukov et al., 2022). Representing the estimation error through independent subject contributions allows us to analyze sparse observations without reconstructing individual trajectories. We derive explicit bounds for Gaussian approximation, conditional bootstrap approximation, and joint coverage error. These bounds quantify how sample size,

dimension, observation frequency, and bandwidth determine inferential accuracy, providing a quantitative explanation of the sparse-to-dense phase transition and distinguishing its consequences for approximation error and band width.

A Gaussian multiplier bootstrap calibrates the joint maximum directly from fitted subject contributions, avoiding separate estimation of the functional-error covariance surface or selection of a sampling regime. We establish its validity while accounting for fitted residuals, estimated design matrices, and studentization. The framework accommodates local polynomial estimators of any fixed degree and yields joint bands for VCMs with random time-varying predictors and growing dimension. We further extend the analysis to high-dimensional functional means, with simultaneous coverage over time and a growing collection of coordinates.

Section 2 introduces the model, estimator, and regularity conditions. Section 3 develops the linear representation, Gaussian and bootstrap approximation, simultaneous bands, and a high-dimensional mean extension. Section 4 reports simulations and an exploratory Mayo PBC analysis. The Supplementary Material provides proofs, additional sampling-frequency and mean results, and bandwidth and implementation details.

## 2. Model and regularity conditions

### 2.1 Model and estimation

Let $Y_i(t)$ be a scalar response and $\mathbf{X}_i(t) = \{X_{i1}(t), \ldots, X_{ip}(t)\}^\intercal$ a vector of predictors for subject $i$ at time $t \in \mathcal{T} = [0, 1]$. We consider the functional varying-coefficient model

$$Y_i(t) = \mathbf{X}_i(t)^\intercal \boldsymbol{\beta}(t) + \varepsilon_i(t), \qquad i = 1, \ldots, n, \tag{2.1}$$

where $\boldsymbol{\beta}(t) = \{\beta_1(t), \ldots, \beta_p(t)\}^\intercal$ is a vector of unknown coefficient functions and $\varepsilon_i(t)$ is a functional error process. The coefficient $\beta_k(t)$ describes the linear association between the response and the $k$th predictor at time $t$, with the other predictors included in the model. A constant coordinate $X_{i1}(t) \equiv 1$ supplies an intercept; time-independent covariates can likewise be included as constant coordinate processes. Both the predictors and their associations with the response may vary over time, and the dimension $p = p_n$ may increase with $n$.

The subject processes are independent across $i$, while dependence within a subject is retained. We impose the pointwise orthogonality condition $\mathbb{E}\{\mathbf{X}_i(t)\varepsilon_i(t)\} = \mathbf{0}$. Writing $\boldsymbol{\Omega}(t) = \mathbb{E}\{\mathbf{X}_i(t)\mathbf{X}_i(t)^\intercal\}$, this condition identifies the population linear regression coefficient as

$$\boldsymbol{\beta}(t) = \boldsymbol{\Omega}(t)^{-1}\mathbb{E}\{\mathbf{X}_i(t)Y_i(t)\}$$

whenever $\mathbf{\Omega}(t)$ is nonsingular. In particular, this formulation does not require the predictor and error processes to be independent.

The trajectories are observed at subject-specific times. Subject $i$ supplies $(T_{ij}, \mathbf{X}_{ij}, Y_{ij})$, $j = 1, \ldots, m_i$, where

$$\mathbf{X}_{ij} = \mathbf{X}_i(T_{ij}), \qquad Y_{ij} = Y_i(T_{ij}), \qquad \varepsilon_{ij} = \varepsilon_i(T_{ij}).$$

The observation times have a common density $f$ and are independent of the subject processes. The counts $m_i$ may differ across subjects and may remain bounded or increase with $n$. We pool observations across subjects for estimation and simultaneous inference over all $t \in \mathcal{T}$ and all $p$ coefficient functions, without first reconstructing individual trajectories.

To construct a local polynomial estimator, fix a degree $r \geq 0$ and a bandwidth $h > 0$. At every target time $t \in \mathcal{T}$, approximate each coefficient function by its Taylor polynomial in a neighborhood of $t$. Under the smoothness condition in Assumption 2,

$$\beta_k(v) = \sum_{a=0}^{r} \frac{h^a \beta_k^{(a)}(t)}{a!} \left( \frac{v-t}{h} \right)^a + O(|v-t|^{r+1}), \qquad |v-t| \leq h,$$

where $v \in \mathcal{T}$, $\beta_k^{(0)} = \beta_k$, and the remainder bound is uniform in $k$ and $t$. Thus the coefficient of the normalized power $\{(v-t)/h\}^a$ approximates $h^a \boldsymbol{\beta}^{(a)}(t)/a!$. The polynomial approximation concerns the coefficient functions; the predictors enter through their observed values $\mathbf{X}_{ij}$.

Let $K$ be a nonnegative kernel supported on $[-1,1]$ and put $K_h(u) = K(u/h)/h$. Define

$$\boldsymbol{\theta}_r(u) = (1, u, \ldots, u^r)^\top, \qquad \boldsymbol{\psi}_{ij}(t) = \boldsymbol{\theta}_r\{(T_{ij} - t)/h\} \otimes \mathbf{X}_{ij},$$

where $\otimes$ denotes the Kronecker product. Write the local polynomial coefficients as a block vector

$$\mathbf{b} = (\mathbf{b}_0^\top, \ldots, \mathbf{b}_r^\top)^\top \in \mathbb{R}^{(r+1)p}.$$

The corresponding local predictor is

$$\boldsymbol{\psi}_{ij}(t)^\top \mathbf{b} = \sum_{a=0}^{r} \mathbf{X}_{ij}^\top \mathbf{b}_a \left(\frac{T_{ij} - t}{h}\right)^a.$$

We estimate the blocks by equal-subject weighted least squares:

$$\widehat{\mathbf{b}}(t) = \arg\min_{\mathbf{b} \in \mathbb{R}^{(r+1)p}} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t)\{Y_{ij} - \boldsymbol{\psi}_{ij}(t)^\top \mathbf{b}\}^2. \tag{2.2}$$

The kernel localizes the fit to observations within distance $h$ of $t$. The factor $1/m_i$ first averages the losses within each subject, so that every subject receives the same total weight before localization. This weighting keeps subjects with more observations from receiving greater total weight solely because of their observation counts.

For computation, define the empirical and population design matrices

$$\widehat{\mathbf{D}}_h(t) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t) \boldsymbol{\psi}_{ij}(t) \boldsymbol{\psi}_{ij}(t)^\top, \qquad \mathbf{D}_h(t) = \mathbb{E}\{\widehat{\mathbf{D}}_h(t)\}.$$

When $\widehat{\mathbf{D}}_h(t)$ is invertible, the normal equations for (2.2) give

$$\widehat{\mathbf{b}}(t) = \widehat{\mathbf{D}}_h(t)^{-1}\frac{1}{n}\sum_{i=1}^{n}\frac{1}{m_i}\sum_{j=1}^{m_i}K_h(T_{ij}-t)\boldsymbol{\psi}_{ij}(t)Y_{ij}.$$

Partition $\widehat{\mathbf{b}}(t)$ into its $p$-dimensional blocks $\widehat{\mathbf{b}}_0(t),\ldots,\widehat{\mathbf{b}}_r(t)$. Evaluating the fitted polynomial at its center gives the coefficient estimator

$$\widehat{\boldsymbol{\beta}}(t) = \widehat{\mathbf{b}}_0(t) = \mathbf{L}\widehat{\mathbf{b}}(t), \qquad \mathbf{L} = [\mathbf{I}_p\ \mathbf{0}\ \cdots\ \mathbf{0}].$$

The same fit also supplies derivative estimators. Differentiating the fitted polynomial with respect to its time argument and evaluating at $t$ yields

$$\widehat{\boldsymbol{\beta}}^{(a)}(t) = \frac{a!}{h^a}\widehat{\mathbf{b}}_a(t), \qquad a = 0,\ldots,r, \tag{2.3}$$

with $\widehat{\boldsymbol{\beta}}^{(0)} = \widehat{\boldsymbol{\beta}}$. For local linear fitting ($r = 1$), the first two blocks estimate $\boldsymbol{\beta}(t)$ and $h\boldsymbol{\beta}'(t)$, respectively, so that $\widehat{\boldsymbol{\beta}}'(t) = \widehat{\mathbf{b}}_1(t)/h$. For $r \geq 2$, the second derivative is estimated by $2\widehat{\mathbf{b}}_2(t)/h^2$. The notation in (2.3) refers to derivatives of the local fitted polynomial at its center; these estimators need not equal derivatives of the curve obtained by moving the fitting center $t$. The simultaneous bands developed below concern the coefficient functions corresponding to $a = 0$.

For a vector or matrix function $A$, $\|A\|_{\infty,I}$ is the largest absolute entry, maximized over $t \in I$; $\|A\|_{2,I}$ is the analogous supremum of its Euclidean norm when $A$ is a vector. Constants in $\lesssim$ and $O_P$ are independent of $n, p, h$

and the counts. We use $\mathcal{F}_n$ for the sigma-field generated by all observations; conditional probabilities and expectations are written explicitly given $\mathcal{F}_n$.

### 2.2 Regularity conditions

The dimension $p = p_n$ may grow; $r$ and $q \geq 4$ are fixed. All constants in the following conditions are uniform in $n$, $p$, and the observation counts. We use $\|\cdot\|_{\mathrm{op}}$ for the spectral norm and $\preceq$ for the positive-semidefinite order. All processes are assumed jointly measurable and separable, and all suprema below are measurable. All time-indexed conditions hold uniformly over $\mathcal{T} = [0, 1]$.

**Assumption 1** (Sampling)**.** *For each $n$, the pairs $(\mathbf{X}_i, \varepsilon_i)$ are independent and identically distributed across subjects, and the counts $m_i \geq 1$ are deterministic. Their common process law may depend on $n$. The times $T_{ij}$ are mutually independent with common density $f$, and independent of the subject processes. The model* (2.1) *satisfies $\mathbb{E}\{\mathbf{X}(t)\varepsilon(t)\} = \mathbf{0}$ for all $t$.*

**Assumption 2** (Smoothness and identification)**.** *The coordinates of $\boldsymbol{\beta}$ belong to a common bounded $C^{r,1}(\mathcal{T})$ class: they have $r$ continuous derivatives, and their $r$th derivatives are Lipschitz; for $r = 0$, the functions themselves are Lipschitz. The density $f$ is Lipschitz and $0 < c_f \leq f \leq C_f$. With*

$\mathbf{\Omega}(t) = \mathbb{E}\{\mathbf{X}(t)\mathbf{X}(t)^\intercal\}$,

$$c_\Omega \mathbf{I}_p \preceq \mathbf{\Omega}(t) \preceq C_\Omega \mathbf{I}_p, \qquad \|\mathbf{\Omega}(t) - \mathbf{\Omega}(u)\|_{\mathrm{op}} \le C|t-u|.$$

**Assumption 3** (Kernel and bandwidth)**.** *The kernel $K$ is nonnegative, symmetric, Lipschitz on $\mathbb{R}$, supported on $[-1,1]$, and has integral one. The bandwidth is deterministic, $0 < h \le 1/2$, and $h \to 0$.*

**Assumption 4** (Coordinate moments)**.** *For a fixed $q \ge 4$,*

$$\max_{k\le p} \mathbb{E}\{\sup_t |X_k(t)|^{2q}\} + \max_{k\le p} \mathbb{E}\{\sup_t |X_k(t)\varepsilon(t)|^q\} \le C.$$

*Write*

$$F_i = 1 + \sup_t \|\mathbf{X}_i(t)\|_\infty^2 + \sup_t \|\mathbf{X}_i(t)\varepsilon_i(t)\|_\infty, \qquad \|F_i\|_{L^q} \le K_n.$$

*The coordinate moment bounds yield the worst-case upper bound $\|F_i\|_{L^q} \le Cp^{1/q}$, and we can take $1 \le K_n \le Cp^{1/q}$. Stronger assumptions that yield smaller $K_n$ lead to sharper rates.*

**Assumption 5** (Score covariance)**.** *Define the score covariance kernel by*

$$\mathbf{C}(u,v) = \mathbb{E}\{\mathbf{X}(u)\varepsilon(u)\varepsilon(v)\mathbf{X}(v)^\intercal\}.$$

*It satisfies*

$$\sup_{u,v} \|\mathbf{C}(u,v)\|_{\mathrm{op}} \le C,$$

$$\|\mathbf{C}(u,v) - \mathbf{C}(u',v')\|_{\mathrm{op}} \le C(|u-u'| + |v-v'|).$$

*For $\mathbf{V}_F(t) = \mathbf{\Omega}(t)^{-1}\mathbf{C}(t,t)\mathbf{\Omega}(t)^{-1}$, $\inf_{t\in\mathcal{T}} \min_{k\le p} [\mathbf{V}_F(t)]_{kk} \ge c_F > 0$.*

Assumptions 1–3 describe independently sampled subjects with noninformative random observation times, smooth coefficients, and a uniformly well-conditioned design, while retaining within-subject dependence. Examples include well-conditioned banded second-moment matrices with uniformly Lipschitz entries and a fixed number of nonzero diagonals, and the Epanechnikov kernel used in our numerical studies.

Assumptions 4–5 accommodate growing dimension, polynomial tails of sufficiently high order, and dependence across times and coordinates, with $K_n$ recording the size of the largest-coordinate envelope. The coordinate variance condition holds, for example, for mean-zero functional errors independent of the predictors with uniformly positive variances. The general condition also allows singular score covariance matrices.

## 3. Simultaneous inference

The argument follows three steps: a subject-level linear representation, Gaussian and bootstrap approximation, and calibration of the joint bands.

### 3.1 Linear representation

Define the subject score and its population-weighted contribution by

$$\mathbf{u}_i(t) = \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t) \boldsymbol{\psi}_{ij}(t) \varepsilon_{ij}, \qquad \boldsymbol{\Psi}_i(t) = \mathbf{L}\mathbf{D}_h(t)^{-1} \mathbf{u}_i(t).$$

The $\mathbf{\Psi}_i$ are independent and centered. Put $s = r + 1$ and

$$\kappa_{a,n} = \frac{1}{n}\sum_{i=1}^{n} m_i^{-a} \quad (a = 1, 3), \qquad v_n^2 = 1 + \frac{\kappa_{1,n}}{h}, \qquad L_n = \log(np/h),$$
$$\delta_n = v_n\sqrt{\frac{L_n}{n}} + \frac{K_n L_n}{n^{1-1/q}h},$$
$$\rho_n = p^{q/(2q-2)} v_n\sqrt{\frac{L_n}{n}} + \frac{pL_n}{n^{1-1/q}h}. \tag{3.1}$$

Here $K_n \geq 1$ bounds the $L^q$ norm of the largest coordinate design and score envelope, defined in Assumption 4. The scale $v_n^2$ is the order of the average coordinate variance of the subject contributions. The rate $\delta_n$ controls score coordinates, whereas $\rho_n$ controls the empirical design in spectral norm. Direct matrix moment bounds give $\rho_n \lesssim p\delta_n$. The matrix envelope has $L^q$ norm of order at most $p$, directly from the coordinate moments.

**Theorem 1** (Uniform linear representation)**.** *Under Assumptions 1–4, if $\rho_n \to 0$, the empirical design is uniformly invertible on $\mathcal{T}$ with probability tending to one, and*

$$\widehat{\boldsymbol{\beta}}(t) - \boldsymbol{\beta}(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{\Psi}_i(t) + \mathbf{R}_n(t), \qquad t \in \mathcal{T},$$

*where*

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_{2,\mathcal{T}} = O_P\{\sqrt{p}(\delta_n + h^s)\},$$
$$\|\mathbf{R}_n\|_{\infty,\mathcal{T}} = O_P\{\sqrt{p}(\rho_n\delta_n + h^s)\}.$$

The remainder records both estimation of the local design and smoothing bias. Its bound over the full fitting interval also controls residuals evaluated at the observed times. When the variance terms dominate $\rho_n$ and $\delta_n$, the stochastic remainder has order $p^{1+1/(2q-2)}v_n^2 L_n/n$. The factor $\sqrt{p}$ in the bias bound reflects coordinatewise smoothness of the $p$ coefficient functions.

### 3.2 Gaussian and bootstrap approximation

Let $\tau_k^2(t) = n^{-1}\sum_i \mathbb{E}\{\Psi_{ik}^2(t)\}$ and let $G_n$ be a centered continuous Gaussian process with

$$\mathbb{E}\{G_{n,k}(u)G_{n,l}(t)\} = \frac{\sum_{i=1}^n \mathbb{E}\{\Psi_{ik}(u)\Psi_{il}(t)\}}{n\tau_k(u)\tau_l(t)}.$$

Under Assumption 5, $\tau_k(t) \asymp v_n$ uniformly; the covariance calculation is in Section A.2. For comparison with $S_{G,n} = \|G_n\|_{\infty,\mathcal{T}}$, define

$$T_n^\circ = \sup_{t\in\mathcal{T}} \max_{k\le p} \frac{\sqrt{n}\,|\widehat{\beta}_k(t) - \beta_k(t)|}{\tau_k(t)}.$$

Set $T_n^\circ = 0$ on the exceptional design event. Define the moment scales and the Gaussian approximation error by

$$\begin{aligned}
\Lambda_n &= \frac{1+\kappa_{3,n}/h^3}{v_n^4}, \qquad B_n = \max\left\{\frac{\sqrt{p}\,K_n}{hv_n}, p\sqrt{\Lambda_n}\right\},\\
\gamma_n &= \left(\frac{B_n^2 L_n^5}{n}\right)^{1/4} + \frac{B_n L_n^{3/2-1/q}}{n^{1/2-1/q}} \\
&\quad + \left(\frac{\sqrt{npL_n}\,\rho_n\delta_n}{v_n}\right)^{1/2} + \frac{\sqrt{npL_n}\,h^s}{v_n}.
\end{aligned} \tag{3.2}$$

**Theorem 2** (Gaussian approximation)**.** *Under Assumptions 1–5, if* $\rho_n \to 0$*, then, for all sufficiently large* $n$*,*

$$\sup_z |\mathbb{P}(T_n^\circ \le z) - \mathbb{P}(S_{G,n} \le z)| \le C\gamma_n.$$

The first two terms of $\gamma_n$ control Gaussian approximation of the subject scores; the remaining terms account for design estimation and smoothing bias. Smaller discretization and exceptional-event terms are absorbed into this bound, as shown in the proof.

For implementation, form $\widehat{\varepsilon}_{ij} = Y_{ij} - \mathbf{X}_{ij}^\intercal \widehat{\boldsymbol{\beta}}(T_{ij})$ and

$$\begin{aligned} \widehat{\boldsymbol{\Psi}}_i(t) &= \mathbf{L}\widehat{\mathbf{D}}_h(t)^{-1} \frac{1}{m_i} \sum_{j=1}^{m_i} K_h(T_{ij} - t)\boldsymbol{\psi}_{ij}(t)\widehat{\varepsilon}_{ij}, \\ \widehat{\sigma}_k^2(t) &= \frac{1}{n}\sum_{i=1}^{n} \widehat{\Psi}_{ik}^2(t). \end{aligned} \tag{3.3}$$

Independently of the data, draw independent $\xi_i \sim N(0,1)$ and set

$$G_{n,k}^*(t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \xi_i \frac{\widehat{\Psi}_{ik}(t)}{\widehat{\sigma}_k(t)}, \qquad S_n^* = \|G_n^*\|_{\infty,\mathcal{T}}. \tag{3.4}$$

A single multiplier is shared by all observations and coordinates of a subject. The statistic being calibrated is

$$T_n = \sup_{t\in\mathcal{T}} \max_{k\le p} \frac{\sqrt{n}\,|\widehat{\beta}_k(t) - \beta_k(t)|}{\widehat{\sigma}_k(t)}.$$

Collect the covariance and fitted-score errors in

$$e_n = p\left\{ \sqrt{\frac{\Lambda_n L_n}{n}} + \frac{K_n^2 L_n}{n^{1-2/q}h^2 v_n^2} + \sqrt{p}(\delta_n + h^s) \right\}. \tag{3.5}$$

On samples with a singular local design or a nonpositive fitted variance, set $T_n, S_n^*, G_n^*$ to zero and the bands to the empty set. The oracle statistic $T_n^\circ$ keeps its earlier convention. The proofs include the probability of these exceptional samples. The first two terms in $e_n$ control empirical covariance fluctuations;

This bound uses the same observations for estimation and calibration.

**Theorem 3** (Bootstrap approximation)**.** *Under Assumptions 1–5, if* $e_n \to 0$*, then*

$$\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - \mathbb{P}(S_{G,n} \le z)| = O_P(\sqrt{e_n}\, L_n),$$

$$\sup_z |\mathbb{P}(T_n \le z) - \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n)| = O_P(\gamma_n + \sqrt{e_n}\, L_n).$$

The bounds are consistent if $\gamma_n \to 0$ and $e_n L_n^2 \to 0$. These conditions specify the admissible joint growth of dimension and sampling frequency, together with the bandwidth; they do not treat $p$ as a constant.

### 3.3 Simultaneous confidence bands

For $v \in (0,1)$, define the conditional quantile

$$\widehat{q}_v = \inf\{z : \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) \ge v\}. \tag{3.6}$$

At confidence level $1-\alpha$, the bands are

$$\mathcal{C}_{1-\alpha,k}(t) = \left[\widehat{\beta}_k(t) \;\pm\; \frac{\widehat{q}_{1-\alpha}\widehat{\sigma}_k(t)}{\sqrt{n}}\right], \qquad t \in \mathcal{T}, \quad k = 1, \dots, p.$$

The common critical value calibrates coverage jointly over time and coefficient functions.

**Theorem 4** (Simultaneous coverage)**.** *Under Assumptions 1–5, suppose* $\gamma_n \to 0$ *and* $e_n L_n^2 \to 0$*. For every fixed* $\alpha \in (0,1)$*,*

$$\left|\mathbb{P}\{\beta_k(t) \in \mathcal{C}_{1-\alpha,k}(t) \text{ for all } t \in \mathcal{T},\ k \le p\} - (1-\alpha)\right| \le C\{\gamma_n + (e_n L_n^2)^{1/3}\}.$$

The bands use the uncorrected local polynomial estimate, so the bandwidth must make smoothing bias negligible for simultaneous inference. In particular, $\gamma_n \to 0$ includes the undersmoothing requirement

$$\frac{\sqrt{npL_n}\, h^{r+1}}{v_n} \to 0. \tag{3.7}$$

The uniform bias bound is $O(\sqrt{p}\, h^{r+1})$, and the coordinate standard errors have order $v_n/\sqrt{n}$. The additional factor $\sqrt{L_n}$ controls the effect of this bias on the distribution of the joint maximum. The remaining terms of $\gamma_n$ and $e_n$ also restrict how small $h$ may be. The multiplier bootstrap calibrates stochastic fluctuations; the displayed bandwidth condition controls smoothing bias.

The coverage bound follows from a probability bound for covariance error and a bracket for the random critical value. In practice, $\widehat{q}_{1-\alpha}$ is approximated by the empirical quantile of independent multiplier draws.

Section C.3 describes simulation and grid error. The supplementary results also sharpen the rates under comparable counts.

### 3.4 Phase transition for inference

To display the role of repeated measurements, take $p$ fixed and $m_i = m = m_n$ in this subsection. Then $K_n = O(1)$ and the coordinate standard errors have order $\{n^{-1} + (nmh)^{-1}\}^{1/2}$. The following specialization describes the transition through both Gaussian approximation and band width.

**Corollary 1** (Gaussian approximation and band width)**.** *Under Assumptions 1–5, take fixed $p$, $m_i = m$, and $q > 4$. Suppose $h \geq n^{-A}$ for some fixed $A > 0$, and*

$$\begin{aligned} n \wedge (nmh) &\gtrsim n^{4/q}(\log n)^{3-4/q}, \\ \{n \wedge (nmh)\}^{3/4} h^s &\lesssim (\log n)^{5/4}. \end{aligned} \tag{3.8}$$

*Then, for all sufficiently large $n$,*

$$\begin{aligned} &\sup_z |\mathbb{P}(T_n^\circ \leq z) - \mathbb{P}(S_{G,n} \leq z)| \\ &\qquad \leq C \left\{ \frac{\log^7 n}{n \wedge (nmh)} \right\}^{1/4}. \end{aligned} \tag{3.9}$$

*For every fixed $\alpha \in (0,1)$, the bands have simultaneous coverage tending to $1-\alpha$. Their half-widths $w_{n,k}(t) = \widehat{q}_{1-\alpha}\widehat{\sigma}_k(t)/\sqrt{n}$ satisfy, with probability*

*tending to one,*

$$c\left\{\frac{1}{\sqrt{n}}+\sqrt{\frac{\log(1/h)}{nmh}}\right\}\le w_{n,k}(t)\le C\left\{\frac{1}{\sqrt{n}}+\sqrt{\frac{\log(1/h)}{nmh}}\right\},\qquad t\in\mathcal{T},\qquad k\le p. \tag{3.10}$$

The first balance in (3.8) makes the finite-moment term no larger than the Gaussian approximation term; the second places smoothing bias below the same bound. These are sufficient simplifications of the general rates. They hold, for example, for $q = 8$, $r \ge 1$, and $h = n^{-2/5}$ for every $m \ge 1$. Section C.1 gives the derivation and verifies the coverage conditions.

From the Gaussian approximation perspective, $mh \le 1$ gives the bound $\{\log^7 n/(nmh)\}^{1/4}$, whereas $mh \ge 1$ gives $(\log^7 n/n)^{1/4}$. Thus the effective sample size increases with the number $nmh$ of observations in local windows and reaches the order of the number $n$ of independent subjects at $mh \asymp 1$. The reference Gaussian law retains the corresponding covariance throughout this transition.

Band width gives a complementary perspective. When $mh \ll \log(1/h)$, the half-width has order $\{\log(1/h)/(nmh)\}^{1/2}$; when $mh \gtrsim \log(1/h)$, it has order $n^{-1/2}$. The additional logarithm is the cost of simultaneous coverage over time. Consequently, the crossover for band width occurs later than the crossover for coordinate variance.

Repeated measurements reduce the uncertainty from observing a trajectory at finitely many times. They continue to share the same subject-specific functional error, however, so increasing their number cannot remove variation between subjects. Once the local sampling contribution is small, further visits mainly resolve an already observed trajectory and yield diminishing improvements in inference. Increasing the number of independent subjects reduces the remaining uncertainty and narrows the bands beyond this plateau.

### 3.5 A mean-function corollary

The same subject-score argument applies to a growing collection of mean functions. Replace the random predictor vector by a scalar intercept in each of $p = p_n$ response coordinates:

$$Y_{ij,k} = \mu_k(T_{ij}) + \varepsilon_{i,k}(T_{ij}), \qquad j \le m_i, \quad k \le p = p_n.$$

The mean-zero error processes are independent and identically distributed across subjects and independent of the observation times; their coordinates may be dependent. The counts $m_i$ are deterministic positive integers, and all observation times are mutually independent with common density $f$. Keep the sampling-density, kernel, and bandwidth conditions from Section 2.2, and suppose the $\mu_k$ belong to a common bounded $C^{r,1}(\mathcal{T})$ class.

Retain polynomial moment assumptions: for fixed $q \geq 4$,

$$\begin{aligned} \max_{k\leq p} \mathbb{E}\{\sup_t |\varepsilon_{i,k}(t)|^q\} \leq C, \\ \left\| 1 + \sup_t \max_{k\leq p} |\varepsilon_{i,k}(t)| \right\|_{L^q} \leq K_n, \qquad 1 \leq K_n \leq Cp^{1/q}. \end{aligned} \tag{3.11}$$

The covariance entries $C_{kl}(u,t) = \mathbb{E}\{\varepsilon_{i,k}(u)\varepsilon_{i,l}(t)\}$ are jointly Lipschitz with one common constant, and $\inf_{t,k} C_{kk}(t,t) \geq c > 0$. These are coordinatewise conditions; the full covariance may be singular.

Fit each mean by the equal-subject local polynomial criterion (2.2), using scalar predictor one and response $Y_{ij,k}$. All $p$ fits share the same $(r+1)\times(r+1)$ matrix

$$\widehat{H}(t) = \frac{1}{n}\sum_i \frac{1}{m_i}\sum_j K_h(T_{ij}-t)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)^{\intercal}.$$

Let $\widehat{\mu}_k$ denote the fitted intercept, and put $\widehat{\varepsilon}_{ij,k} = Y_{ij,k} - \widehat{\mu}_k(T_{ij})$. With $\mathbf{e}_0 = (1,0,\ldots,0)^{\intercal} \in \mathbb{R}^{r+1}$, the fitted subject contributions are

$$\widehat{\Psi}_{ik}(t) = \mathbf{e}_0^{\intercal}\widehat{H}(t)^{-1}\frac{1}{m_i}\sum_j K_h(T_{ij}-t)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)\widehat{\varepsilon}_{ij,k}.$$

Their oracle versions use $H = \mathbb{E}\widehat{H}$ and the true errors. Define $\tau_k$, $\widehat{\sigma}_k$, and the subject multiplier bootstrap as before, with one multiplier shared across all coordinates of a subject. In this subsection, $T_n^{\circ}, T_n, S_n^*, S_{G,n}$ denote the corresponding mean-model statistics, and the bands $\mathcal{C}_{1-\alpha,k}$ are centered at $\widehat{\mu}_k$, with the same exceptional-sample conventions. The scales $v_n, L_n$ remain those in (3.1), and $\delta_n$ uses the mean envelope $K_n$ in (3.11).

Only the design-concentration and coordinate-mixing steps change. Define the scalar-design rate and the resulting approximation errors by

$$\begin{aligned} a_n &= \sqrt{\frac{\kappa_{1,n}\log(n/h)}{nh}} + \frac{\log(n/h)}{nh}, \\ \gamma_n^{\mu} &= \left(\frac{K_n^2 L_n^5}{nh^2 v_n^2}\right)^{1/4} + \frac{K_n L_n^{3/2-1/q}}{n^{1/2-1/q} h v_n} \\ &\quad + \left(\frac{\sqrt{nL_n}\, a_n \delta_n}{v_n}\right)^{1/2} + \frac{\sqrt{nL_n}\, h^s}{v_n}, \\ e_n^{\mu} &= \sqrt{\frac{\Lambda_n L_n}{n}} + \frac{K_n^2 L_n}{n^{1-2/q} h^2 v_n^2} + \delta_n + h^s. \end{aligned} \tag{3.12}$$

The shared design has fixed size, its inverse acts separately on each coordinate, and each residual uses only its own mean estimate. Thus $a_n$ replaces the regression design rate $\rho_n$, and the polynomial factors in $p$ from coordinate mixing and residual fitting disappear. The score rate $\delta_n$ and covariance scale $\Lambda_n$ retain their definitions, with the actual number of responses $p$ in $L_n$.

**Corollary 2** (Joint mean inference under polynomial moments)**.** *Under the preceding conditions, suppose $a_n \to 0$. Then, uniformly in $t, k$,*

$$\widehat{\mu}_k(t) - \mu_k(t) = \frac{1}{n}\sum_i \Psi_{ik}(t) + R_{nk}(t),$$

$$\|\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|_{\infty,\mathcal{T}} = O_P(\delta_n + h^s), \qquad \|\mathbf{R}_n\|_{\infty,\mathcal{T}} = O_P(a_n\delta_n + h^s).$$

*Here $\mathbf{R}_n = (R_{n1}, \ldots, R_{np})^{\intercal}$. For all sufficiently large $n$,*

$$\sup_z |\mathbb{P}(T_n^{\circ} \le z) - \mathbb{P}(S_{G,n} \le z)| \le C\gamma_n^{\mu}. \tag{3.13}$$

*If* $e_n^\mu \to 0$*, the bootstrap conclusions are*

$$\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - \mathbb{P}(S_{G,n} \le z)| = O_P(\sqrt{e_n^\mu}\, L_n),$$

$$\sup_z |\mathbb{P}(T_n \le z) - \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n)| = O_P(\gamma_n^\mu + \sqrt{e_n^\mu}\, L_n).$$

*If* $\gamma_n^\mu \to 0$ *and* $e_n^\mu L_n^2 \to 0$*, then, for every fixed* $\alpha \in (0,1)$*,*

$$\begin{aligned} &|\mathbb{P}\{\mu_k(t) \in \mathcal{C}_{1-\alpha,k}(t) \text{ for all } t \in \mathcal{T},\ k \le p\} - (1-\alpha)| \\ &\quad \le C\{\gamma_n^\mu + (e_n^\mu L_n^2)^{1/3}\}. \end{aligned} \tag{3.14}$$

The proof specializes the linearization, covariance, and quantile-bracketing arguments of Theorems 1–4; details are in Section D. The mean bands also use an uncorrected estimate, so $\gamma_n^\mu \to 0$ requires

$$\frac{\sqrt{nL_n}\, h^{r+1}}{v_n} \to 0. \tag{3.15}$$

This puts the bias below the coordinate standard error by a factor $\sqrt{L_n}$, which suffices for the joint-maximum comparison used here. With $m_i = m$, it is equivalent, up to constants, to $\{n \wedge (nmh)\}L_n h^{2r+2} \to 0$. For local linear fits this becomes $nmh^5 L_n \to 0$ when $mh \to 0$, and $nh^4 L_n \to 0$ when $mh$ is bounded away from zero.

The only dimension dependence is through $L_n$ and the envelope $K_n$. Coordinate polynomial moments give the worst-case bound $K_n \lesssim p^{1/q}$; a bounded or polylogarithmic envelope yields logarithmic dimension dependence without requiring exponential moments. In particular, with $q = 8$,

$K_n = O(1)$, local linear fits, $m_i = 1$, and $h = n^{-1/4}$, $\log p = o(n^{3/20})$ gives coverage error $O(n^{-1/8} L_n^{5/6})$. Section D.1 gives a polynomial-tailed model with a dimension-free envelope, demonstrating exponentially many mean functions under sparse sampling.

## 4. Numerical studies

### 4.1 Simulation design and competing methods

The factorial design compares nominal 95% bands with $n \in \{200, 400\}$ subjects, $m \in \{5, 20, 80\}$ observations per subject, and Gaussian, Laplace, or adjusted chi-square innovations in the functional errors. We refer to these three sampling frequencies as sparse, mild, and dense; these are finite-sample labels. Every configuration uses $R = 1{,}000$ independent datasets. We examine a single coefficient in a two-coefficient model, simultaneous coverage of all five coefficients in a five-coefficient model, and the intercept-only mean problem.

Observation times are independent Uniform$(0, 1)$. Predictors and the single functional error are generated from smooth finite-basis processes. Error innovations have mean zero and variance one: $N(0, 1)$, Laplace$(0, 1/\sqrt{2})$, or $(\chi^2_5 - 5)/\sqrt{10}$. The latter two examine heavier symmetric tails and positive skewness, respectively. The standardized $\chi^2_5$ law

has skewness $\sqrt{8/5}$ and excess kurtosis $12/5$, providing a moderately skewed case. The coefficient functions include linear, oscillating, quadratic, and exponential trends. Section E.1 gives the complete generating mechanism and targets. We use local linear fits with the Epanechnikov kernel, one common bandwidth for the estimate and residuals, and $B = 1{,}999$ Gaussian multiplier draws. Both methods in each comparison use the same 201 equally spaced evaluation points in $[0.15, 0.85]$. Coverage probability (CP) requires coverage of the entire grid and all targeted coefficients; average width (AW) averages over that grid and those coefficients. These are grid-coverage measurements.

For coefficient bands, we compare with Cao, Liu and Zhou (2018), abbreviated CLZ. Both local linear estimators use the same bandwidth; with equal counts their point estimates agree, while covariance estimation and calibration differ. For the five-function target, CLZ uses Bonferroni-adjusted marginal bands. Section E.2 specifies its kernel variance estimates and finite-observation Gaussian calibration. The CLZ theory concerns sparse longitudinal sampling, so its mild and dense entries assess performance beyond that regime.

For the mean problem, we compare with the B-spline bands of Cai and Hu (2025), which also accommodate varying sampling frequencies. Their

cubic-spline fit uses BIC-selected knots and a subject-score covariance estimate. The design, full results, and computational details for this specialization are in Section E.

### 4.2 Coverage and width

Table 1 reports joint coverage and width for every sampling frequency and all three innovation distributions. Table 3 in the supplement gives the single-coefficient comparison. At $R = 1{,}000$, the Monte Carlo standard error of CP near 0.95 is approximately 0.007. Complete replication records, configuration-specific uncertainty, and paired coverage differences are supplied with the code. All 54,000 replications of this factorial experiment completed without a singular-fit or variance failure.

The proposed single-coefficient CP ranges from 0.908 to 0.948, compared with 0.052 to 0.773 for CLZ. For joint inference the ranges are 0.822–0.940 and 0.000–0.618, respectively. The proposed coverage is closer to 0.95 in all 36 coefficient-model configurations. Comparing $m = 5$ with $m = 80$ at each fixed target, subject sample size, and error law, both CLZ width and coverage decrease in all twelve comparisons.

For the proposed joint bands, CP at $n = 200$ ranges from 0.822 to 0.914. At $n = 400$ the corresponding range is 0.885–0.940. These ranges

Table 1: Coverage probability (CP) and average width (AW) of nominal 95% bands for all five coefficients jointly. Each row uses 1,000 datasets; $m = 5, 20, 80$ denote sparse, mild, and dense sampling.

| | | | Proposed | | CLZ (Bonferroni) | |
|---|---|---|---|---|---|---|
| Innovations | $n$ | $m$ | CP | AW | CP | AW |
| Gaussian | 200 | 5 | 0.822 | 0.703 | 0.458 | 0.566 |
| | 200 | 20 | 0.906 | 0.545 | 0.042 | 0.288 |
| | 200 | 80 | 0.912 | 0.497 | 0.000 | 0.144 |
| | 400 | 5 | 0.910 | 0.556 | 0.581 | 0.458 |
| | 400 | 20 | 0.919 | 0.412 | 0.073 | 0.229 |
| | 400 | 80 | 0.940 | 0.362 | 0.000 | 0.115 |
| Laplace | 200 | 5 | 0.871 | 0.698 | 0.491 | 0.565 |
| | 200 | 20 | 0.904 | 0.543 | 0.047 | 0.288 |
| | 200 | 80 | 0.914 | 0.492 | 0.000 | 0.143 |
| | 400 | 5 | 0.904 | 0.550 | 0.618 | 0.454 |
| | 400 | 20 | 0.925 | 0.410 | 0.063 | 0.228 |
| | 400 | 80 | 0.929 | 0.361 | 0.001 | 0.115 |
| Adjusted $\chi^2_5$ | 200 | 5 | 0.847 | 0.698 | 0.484 | 0.563 |
| | 200 | 20 | 0.901 | 0.545 | 0.044 | 0.289 |
| | 200 | 80 | 0.910 | 0.494 | 0.000 | 0.144 |
| | 400 | 5 | 0.885 | 0.553 | 0.580 | 0.457 |
| | 400 | 20 | 0.924 | 0.410 | 0.064 | 0.229 |
| | 400 | 80 | 0.934 | 0.362 | 0.000 | 0.115 |

make the remaining finite-sample coverage error explicit. Both Gaussian and non-Gaussian results are reported without selecting configurations by outcome.

For the mean comparison in Table 4, proposed CP is 0.919–0.954, while Cai–Hu CP is 0.932–0.958. Cai–Hu bands are narrower in all 18 mean configurations. Coverage and width should be interpreted together under the specified tuning parameters.

For Gaussian joint bands, increasing $m$ from 20 to 80 changes CP from 0.906 to 0.912 at $n = 200$, and from 0.919 to 0.940 at $n = 400$. The two CP differences have Monte Carlo standard errors 0.013 and 0.011, respectively. Average widths decrease from 0.545 to 0.497 and from 0.412 to 0.362. The smoothing bandwidth is unchanged across $m$ at each fixed $n$ under the prespecified rule. Coverage reflects calibration error and smoothing bias as well as variance, so CP and width need not change in parallel.

### 4.3 Multivariate functional means

We examine Corollary 2 with $n = 200, 400$ subjects, $m = 5, 20, 80$ observations per subject, and $p = 50, 100$ mean functions. The functional errors combine common and coordinate-specific innovations from the same three distributions as above, with cross-coordinate correlation 1/2. Each

$(n, m, \text{innovation law})$ uses 1,000 independent datasets and 999 subject multiplier draws. The two dimensions are nested coordinate prefixes of the same datasets and use the same multipliers. All 36 configurations are evaluated under this fixed design. Coverage requires all coordinates at all 51 equally spaced points on $[0, 1]$, including endpoints. The shared scalar design permits computation in response blocks while retaining a single joint bootstrap critical value. Section E.5 specifies the curves, random streams, bandwidth, implementation checks, and timing convention.

Table 2 reports joint grid coverage and width for every configuration. Monte Carlo uncertainty and computation times are retained with the replication records. Across the fixed design, CP ranges from 0.782 to 0.947. At $n = 400$, $m = 80$, and $p = 100$, CP is 0.935, 0.931, and 0.928 for Gaussian, Laplace, and adjusted chi-square innovations, respectively. Joint undercoverage remains in some configurations. These results concern the stated finite grid and prespecified bandwidth.

Table 2: Joint grid coverage (CP) and average width (AW) for multivariate functional means with $n = 200, 400$, $m = 5, 20, 80$, and $p = 50, 100$. All configurations are reported. Each cell uses 1000 datasets and 999 subject multipliers on 51 points spanning $[0, 1]$.

| | | | Gaussian | | Laplace | | Adjusted $\chi^2_5$ | |
|---|---|---|---|---|---|---|---|---|
| $n$ | $m$ | $p$ | CP | AW | CP | AW | CP | AW |
| 200 | 5 | 50 | 0.860 | 0.927 | 0.878 | 0.927 | 0.802 | 0.926 |
| 200 | 5 | 100 | 0.842 | 0.961 | 0.855 | 0.961 | 0.782 | 0.959 |
| 200 | 20 | 50 | 0.907 | 0.692 | 0.920 | 0.693 | 0.883 | 0.690 |
| 200 | 20 | 100 | 0.904 | 0.719 | 0.920 | 0.719 | 0.877 | 0.717 |
| 200 | 80 | 50 | 0.911 | 0.605 | 0.929 | 0.603 | 0.913 | 0.604 |
| 200 | 80 | 100 | 0.915 | 0.630 | 0.927 | 0.628 | 0.903 | 0.629 |
| 400 | 5 | 50 | 0.924 | 0.716 | 0.925 | 0.715 | 0.865 | 0.714 |
| 400 | 5 | 100 | 0.909 | 0.741 | 0.908 | 0.740 | 0.848 | 0.740 |
| 400 | 20 | 50 | 0.933 | 0.515 | 0.930 | 0.514 | 0.932 | 0.515 |
| 400 | 20 | 100 | 0.925 | 0.534 | 0.921 | 0.534 | 0.921 | 0.534 |
| 400 | 80 | 50 | 0.947 | 0.439 | 0.942 | 0.439 | 0.922 | 0.439 |
| 400 | 80 | 100 | 0.935 | 0.457 | 0.931 | 0.457 | 0.928 | 0.457 |

### 4.4 Mayo PBC sequential biomarker data

We analyze the Mayo Clinic PBC sequential data available as `pbcseq` in the R package `survival` (Survival package authors, 2026), originating from the D-penicillamine trial (Dickson et al., 1985). The full file contains 1,945 visits from 312 patients. Restricting follow-up to 3,000 days and retaining complete observations with positive bilirubin and alkaline phosphatase leaves 1,741 visits from all 312 patients for the regression. Counts range from 1 to 10, with harmonic mean 3.489. Section E shows the observation schedule, the separate mean analysis, and preprocessing details.

The protocol included scheduled visits, with additional visits potentially prompted by worsening health; laboratory measurements can also be missing (Survival package authors, 2026). Consequently, independent observation times and noninformative follow-up are not established. We interpret the fitted bands as an exploratory application under a working sampling model and describe observed-visit associations.

Let $t = \text{day}/3000$, let $D_i$ indicate D-penicillamine, and standardize edema, albumin, prothrombin time, and log alkaline phosphatase by their observation-level means and standard deviations, obtaining $E, A, P, L$.

The regression model is

$$\begin{aligned}\log(\text{bili}_{ij}) = \beta_1(t_{ij}) + \beta_2(t_{ij})D_i + \beta_3(t_{ij})E_{ij} \\ + \beta_4(t_{ij})A_{ij} + \beta_5(t_{ij})P_{ij} + \beta_6(t_{ij})L_{ij} + \varepsilon_i(t_{ij}).\end{aligned} \tag{4.1}$$

For model (4.1), equal-subject weights are recomputed after filtering. The common bandwidth is 0.09510, about 285 days. Calibration uses 9,999 multipliers and is joint over the six coefficients and 201 points from day 450 to day 2,550. The resulting critical value is 3.437; at least 95 patients contribute to each evaluation window.

In Figure 1, the treatment band contains zero throughout the displayed interval, while the log-ALP band remains above zero. Edema and prothrombin time have positive exclusions on portions of follow-up; albumin has short negative exclusions. Bandwidth factors 0.8 and 1.2 and a shorter 2,000-day horizon preserve the treatment and log-ALP descriptions but change the shorter exclusions. Full sensitivity results are in Section E. These are descriptive adjusted associations: conditioning on contemporaneous biomarkers does not identify a randomized treatment effect, and a band containing zero does not establish equivalence.

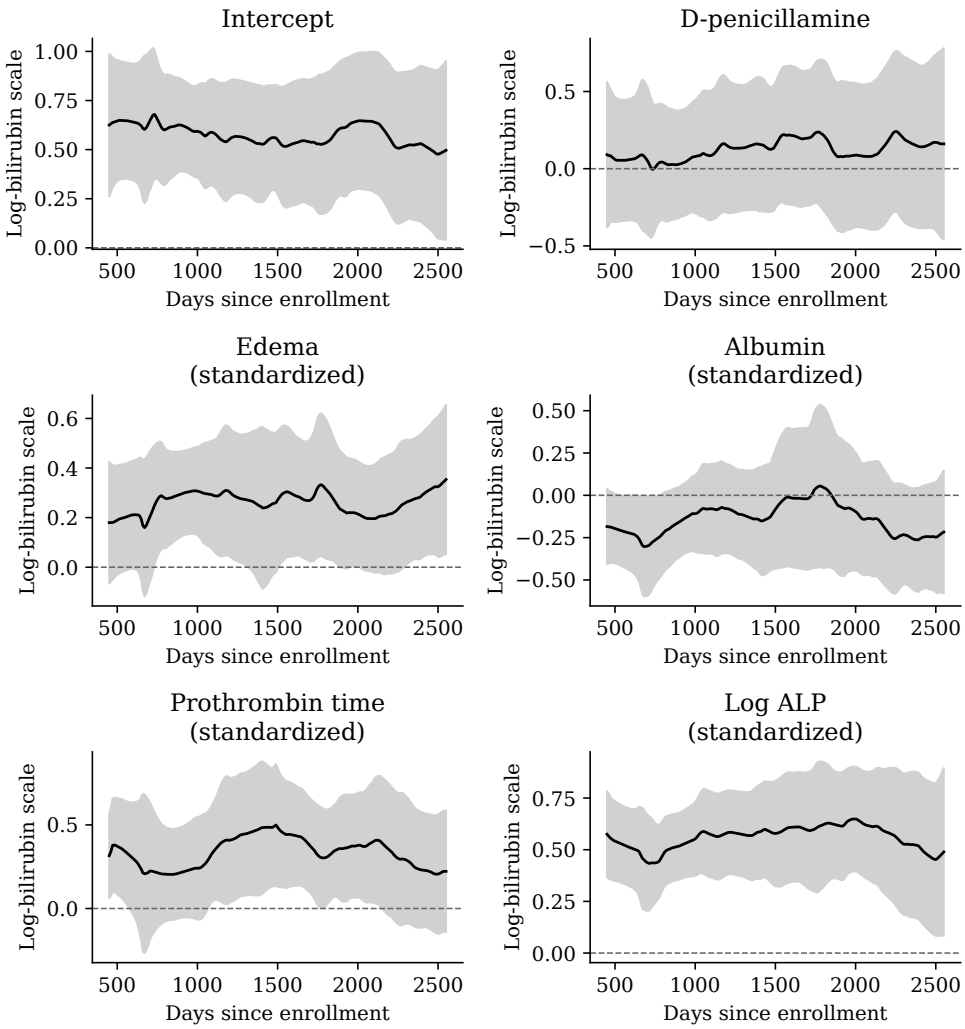



Figure 1: PBC coefficient estimates and nominal joint 95% bands under the working model. Independent visit times and noninformative follow-up are not established for these data.

## 5. Discussion

Subject contributions link local polynomial estimates to Gaussian and bootstrap approximation without requiring individual trajectory reconstruction. The bounds quantify the effects of sampling frequency, dimension, smoothing bias, and estimation of the design and residuals. Their explicit dimension dependence permits growing VCMs; the coordinatewise mean extension has no growing regression matrix to invert. Its dimension dependence enters through a logarithm and the maximum-coordinate moment envelope. Polynomial-tailed models with a bounded envelope allow exponentially many functions under sparse sampling.

The explicit powers of $p$ in our bounds follow worst-case inequalities under the stated assumptions. They account for mixing coordinate scores through the inverse local design and accumulating coefficient errors in fitted residuals. Designs with diagonal second-moment matrices or fixed-size groups of correlated predictors can reduce these dimension factors when the inverse-design absolute row sums remain uniformly bounded. The stated dimension restrictions are therefore sufficient conditions and need not be sharp.

The results assume error-free predictors and observation times independent of the trajectories. They require a bandwidth satisfying the displayed

approximation and bias conditions. The numerical comparisons show that moderate-sample joint undercoverage can remain. Extensions to informative visits, measurement error in predictors, and justified finite-sample corrections are therefore useful directions for further work.

## A. Linearization and covariance bounds

We use Assumptions 1–5 from the main article. Constants are uniform in $n, p, h$ and the observation counts; $r$ and $q$ are fixed, and $s = r + 1$. Write $P = (r + 1)p$ and use $\|\cdot\|_{\max}$ for the largest absolute entry of a vector or matrix at one time point. Estimation, residuals, and inference all use $\mathcal{T} = [0, 1]$. All other notation is as in Sections 2 and 3.

We use the scales in (3.1), (3.2), and (3.5), and the local basis and block selector from Section 2. Set

$$\mathbf{S}_i(t) = m_i^{-1} \sum_j K_h(T_{ij} - t)\boldsymbol{\psi}_{ij}(t)\boldsymbol{\psi}_{ij}(t)^\intercal, \qquad \widehat{\mathbf{D}}_h = n^{-1} \sum_i \mathbf{S}_i,$$

$$\mathbf{u}_i(t) = m_i^{-1} \sum_j K_h(T_{ij} - t)\boldsymbol{\psi}_{ij}(t)\varepsilon_i(T_{ij}), \qquad \bar{\mathbf{u}} = n^{-1} \sum_i \mathbf{u}_i,$$

$$\mathbf{D}_h(t) = \mathbb{E}\{\widehat{\mathbf{D}}_h(t)\}, \qquad \boldsymbol{\Psi}_i(t) = \mathbf{L}\mathbf{D}_h(t)^{-1}\mathbf{u}_i(t).$$

### A.1 Population design and uniform concentration

**Lemma A.1** (Population design)**.** *Under Assumptions 1, 2, and 3, uniformly on $\mathcal{T}$,*

$$c_D\mathbf{I}_P \preceq \mathbf{D}_h(t) \preceq C_D\mathbf{I}_P, \qquad \|\mathbf{D}_h(t) - \mathbf{D}_h(u)\|_{\mathrm{op}} \le Ch^{-1}|t - u|.$$

*Define the kernel-window moment matrix by*

$$I_{t,h} = [-1, 1] \cap [-t/h, (1 - t)/h],$$

$$\boldsymbol{\Gamma}_r(t, h) = \int_{I_{t,h}} \boldsymbol{\theta}_r(z)\boldsymbol{\theta}_r(z)^\intercal K(z)\, dz.$$

*Then, uniformly for $t \in \mathcal{T}$,*

$$\mathbf{D}_h(t) = \mathbf{\Gamma}_r(t,h) \otimes \{f(t)\mathbf{\Omega}(t)\} + O_{\mathrm{op}}(h).$$

*Proof.* The exact formula is

$$\mathbf{D}_h(t) = \int_{I_{t,h}} \boldsymbol{\theta}_r(z)\boldsymbol{\theta}_r(z)^{\intercal} K(z) \otimes \{f(t+hz)\mathbf{\Omega}(t+hz)\}\, dz.$$

For $h \leq 1/2$, this interval contains either $[0,1]$ or $[-1,0]$. Continuity, symmetry, nonnegativity, and unit integral imply that $K$ is positive on a nonempty interval in $(0,1)$ and on its reflection. The polynomial Gram matrix on either half is positive definite: a nonzero polynomial cannot vanish on such an interval. Its dimension $r+1$ is fixed. The eigenvalue bounds for $f\mathbf{\Omega}$ therefore give the asserted uniform lower bound; the upper bound follows from boundedness of the basis and the kernel. The $L^1$ increment, between evaluation points $t$ and $s$, of each scalar kernel $u \mapsto K_h(u-t)((u-t)/h)^a$, $0 \leq a \leq 2r$, is at most $Ch^{-1}|t-s|$. Integrating against the operator-bounded $f\mathbf{\Omega}$ proves the increment bound. Operator-norm Lipschitz continuity of $f\mathbf{\Omega}$ on $\mathcal{T}$ gives the stated expansion with $\mathbf{\Gamma}_r(t,h)$, uniformly in $t$. The inverse matrices have the same uniform operator bound and an $O(h^{-1})$ Lipschitz constant by the resolvent identity. □

For the Taylor remainder, put

$$\mathbf{r}(t,u) = \boldsymbol{\beta}(u) - \sum_{a=0}^{r} \frac{\boldsymbol{\beta}^{(a)}(t)}{a!}(u-t)^a,$$

$$\mathbf{a}_i(t) = m_i^{-1} \sum_j K_h(T_{ij}-t)\boldsymbol{\psi}_{ij}(t)\mathbf{X}_{ij}^{\intercal}\mathbf{r}(t,T_{ij}),$$

and $\mathbf{a}_n = n^{-1}\sum_i \mathbf{a}_i$. Each coordinate of $\mathbf{r}$ is bounded by $Ch^s$ on the kernel support, so $\|\mathbf{r}(t,u)\|_2 \le C\sqrt{p}\,h^s$ there.

**Lemma A.2** (Uniform first and second moments)**.** *Under Assumptions 1–4, define*

$$\mathcal{D}_n = \sup_t \|\widehat{\mathbf{D}}_h(t) - \mathbf{D}_h(t)\|_{\mathrm{op}}, \quad \mathcal{U}_n = \|\bar{\mathbf{u}}\|_{\infty,\mathcal{T}},$$

$$\mathcal{A}_n = \frac{\|\mathbf{a}_n - \mathbb{E}\{\mathbf{a}_n\}\|_{2,\mathcal{T}}}{\sqrt{p}\,h^s}.$$

*Then*

$$\mathbb{E}(\mathcal{D}_n^2 + \mathcal{A}_n^2) \le C\rho_n^2, \qquad \mathbb{E}(\mathcal{U}_n^2) \le C\delta_n^2.$$

*Moreover,*

$$\|\mathbb{E}\{\mathbf{a}_n\}\|_{2,\mathcal{T}} \le C\sqrt{p}\,h^s.$$

*Proof.* We first obtain a matrix variance bound from the existing coordinate moments and the upper eigenvalue bound for $\boldsymbol{\Omega}$. For every $\|\mathbf{v}\|_2 = 1$, Minkowski's inequality gives $\|\mathbf{v}^{\intercal}\mathbf{X}(u)\|_{L^{2q}} \le C\sqrt{p}$, whereas $\|\mathbf{v}^{\intercal}\mathbf{X}(u)\|_{L^2} \le C$. Interpolation and Hölder's inequality therefore imply

$$\mathbb{E}\{X_k(u)^2(\mathbf{v}^{\intercal}\mathbf{X}(u))^2\} \le C\|\mathbf{v}^{\intercal}\mathbf{X}(u)\|^2_{L^{2q/(q-1)}} \le Cp^{1/(q-1)},$$

$$\big\|\mathbb{E}\{\|\mathbf{X}(u)\|_2^2\mathbf{X}(u)\mathbf{X}(u)^{\intercal}\}\big\|_{\mathrm{op}} \le Cp^{q/(q-1)}.$$

These bounds hold uniformly in $u$. The bounded, fixed-dimensional local polynomial basis preserves them up to a constant.

Condition on a subject's trajectories and separate identical from distinct observation indices in the second moments. Identical indices contribute a kernel integral of order $h^{-1}$; distinct indices contribute bounded double integrals. For the latter, the squared conditional mean is bounded by the weighted integral of the squared mark. Explicitly, for a nonnegative weight $w$ and self-adjoint matrix $A(u)$,

$$\left\{\int w(u)A(u)\,du\right\}^2 \preceq \left\{\int w(u)\,du\right\}\int w(u)A(u)^2\,du.$$

Apply this with $w(u) = K_h(u-t)f(u)$ and the matrix mark $\boldsymbol{\theta}_r\boldsymbol{\theta}_r^{\mathsf{T}} \otimes \mathbf{X}_i(u)\mathbf{X}_i(u)^{\mathsf{T}}$; the vector version applies to the Taylor mark. The matrix moment bound and $\|\mathbf{r}(t,u)\|_2 \le C\sqrt{p}\,h^s$ thus give

$$\left\|\mathbb{E}\{(\mathbf{S}_i(t)-\mathbb{E}\mathbf{S}_i(t))^2\}\right\|_{\mathrm{op}} \le Cp^{q/(q-1)}\{1+(m_ih)^{-1}\},$$
$$\frac{\mathbb{E}\|\mathbf{a}_i(t)-\mathbb{E}\mathbf{a}_i(t)\|_2^2}{ph^{2s}} \le Cp^{q/(q-1)}\{1+(m_ih)^{-1}\}.$$

The subject envelopes of $\mathbf{S}_i$ in spectral norm and of $\mathbf{a}_i/(\sqrt{p}\,h^s)$ in Euclidean norm are at most $Ch^{-1}\sup_u\|\mathbf{X}_i(u)\|_2^2$. Minkowski's inequality gives

$$\left\|\sup_u\|\mathbf{X}_i(u)\|_2^2\right\|_{L^q} \le \sum_{k=1}^{p}\left\|\sup_u|X_{ik}(u)|^2\right\|_{L^q} \le Cp.$$

Use a uniform grid $\chi_n$ of mesh at most

$$\Delta_n = n^{-8}h^{10}(pK_n)^{-8},$$

Then $\log(2P|\chi_n|) \le CL_n$. For each subject, form a block diagonal matrix whose blocks are the centered design matrices at all grid points. Its spectral norm is the largest block norm, and the matrix variance parameter of the sum is at most $Cnp^{q/(q-1)}v_n^2$. Its largest summand has $L^2$ norm at most $Cpn^{1/q}/h$. The second-moment inequality in Theorem I of Tropp (2016), divided by $n$, yields

$$\left[\mathbb{E}\max_{t \text{ on grid}} \|\widehat{\mathbf{D}}_h(t) - \mathbf{D}_h(t)\|_{\text{op}}^2\right]^{1/2}$$
$$\le C\left\{p^{q/(2q-2)}v_n\sqrt{L_n/n} + \frac{pL_n}{n^{1-1/q}h}\right\} = C\rho_n.$$

Apply the rectangular version of the same inequality to a block diagonal matrix with column blocks $\{\mathbf{a}_i(t) - \mathbb{E}\mathbf{a}_i(t)\}/(\sqrt{p}\,h^s)$. Both matrix variance parameters are bounded by the displayed vector second moments. This proves the same grid bound for $\mathcal{A}_n$.

For the score, every coordinate has envelope $CF_i/h$ and second moment at most $C\{1 + (m_i h)^{-1}\}$. Applying Lemmas E.1 and E.2(ii) of Chernozhukov, Chetverikov and Kato (2017) simultaneously to the score coordinates and grid points gives expected maximum at most $C\delta_n$ and, above twice that expectation, the tail bound

$$\exp(-cnx^2/v_n^2) + \frac{Cn^{1-q}K_n^q}{h^q x^q}, \qquad x > 0.$$

Integrating, with the polynomial term capped at one, gives the score second moment bound because $q > 2$.

The samplewise Lipschitz constants of the design in spectral norm and of $\mathbf{a}_i/(\sqrt{p}\,h^s)$ in Euclidean norm are at most $CpF_ih^{-2}$; score coordinates have constants $CF_ih^{-2}$. For the Taylor term, use $|\partial_t r_k(t,u)| \le C|u-t|^r$ almost everywhere on the kernel support; for $r = 0$, use the Lipschitz property of $\beta_k$. The compactly supported Lipschitz kernel gives the same increment bound across the support boundary. Thus interpolation errors have $L^2$ norm at most $CpK_n\Delta_n h^{-2}$, which is absorbed by the respective rates. Finally,

$$\mathbb{E}\{\mathbf{a}_n(t)\} = \int K_h(u-t)\{\boldsymbol{\theta}_r((u-t)/h) \otimes \boldsymbol{\Omega}(u)\}\mathbf{r}(t,u)f(u)\,du.$$

The operator bound for $\boldsymbol{\Omega}$, the bounded basis, and $\|\mathbf{r}(t,u)\|_2 \le C\sqrt{p}\,h^s$ prove the last assertion. □

**Lemma A.3** (Uniform linear representation)**.** *Suppose Assumptions 1–4 hold and $\rho_n \to 0$. With probability tending to one, $\widehat{\mathbf{D}}_h$ is invertible throughout $\mathcal{T}$, and*

$$\widehat{\boldsymbol{\beta}}(t) - \boldsymbol{\beta}(t) = n^{-1}\sum_i \boldsymbol{\Psi}_i(t) + \mathbf{R}_n(t), \qquad t \in \mathcal{T}.$$

*Uniformly on this domain,*

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_2 = O_P\{\sqrt{p}(\delta_n + h^s)\}, \qquad \|\mathbf{R}_n\|_{\max} = O_P\{\sqrt{p}(\rho_n\delta_n + h^s)\}.$$

*More precisely, there is an event $\mathcal{E}_n$ with $\mathbb{P}(\mathcal{E}_n^c) \le C\rho_n^2$ such that, on this event,*

$$\sup_t \|\widehat{\mathbf{D}}_h(t)^{-1}\|_{\mathrm{op}} \le C, \qquad \|\widehat{\mathbf{D}}_h^{-1} - \mathbf{D}_h^{-1}\|_{\mathrm{op},\mathcal{T}} \le C\mathcal{D}_n,$$

$$\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_{2,\mathcal{T}} \le C\sqrt{p}(\mathcal{U}_n + h^s), \qquad \|\mathbf{R}_n\|_{\infty,\mathcal{T}} \le C\sqrt{p}\,\mathcal{D}_n\mathcal{U}_n + C\sqrt{p}\,h^s.$$

*Proof.* Let

$$\mathcal{E}_n = \{\mathcal{D}_n \le c_D/2, \quad \mathcal{A}_n \le 1\}.$$

Lemma A.2 and Markov's inequality give its exceptional probability. Weyl's inequality and the resolvent identity give the two inverse bounds. On $\mathcal{E}_n$,

$$\|\mathbf{a}_n\|_{2,\mathcal{T}} \le C\sqrt{p}\,h^s + \sqrt{p}\,h^s\mathcal{A}_n \le C\sqrt{p}\,h^s.$$

The exact normal equation is

$$\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{L}\widehat{\mathbf{D}}_h^{-1}(\bar{\mathbf{u}} + \mathbf{a}_n).$$

Here the Taylor coefficients are stacked with their factors $h^a/a!$, so their first block is exactly $\boldsymbol{\beta}(t)$. Since $\|\bar{\mathbf{u}}\|_2 \le \sqrt{P}\,\mathcal{U}_n$, the estimation bound follows. Subtracting $\mathbf{L}\mathbf{D}_h^{-1}\bar{\mathbf{u}}$ yields

$$\mathbf{R}_n = \mathbf{L}(\widehat{\mathbf{D}}_h^{-1} - \mathbf{D}_h^{-1})\bar{\mathbf{u}} + \mathbf{L}\widehat{\mathbf{D}}_h^{-1}\mathbf{a}_n.$$

Its first term is at most $C\sqrt{p}\,\mathcal{D}_n\mathcal{U}_n$ in Euclidean norm, and hence in maximum norm. The second is at most $C\sqrt{p}\,h^s$ on $\mathcal{E}_n$. Lemma A.2 also gives

$\mathbb{E}(\mathcal{D}_n\mathcal{U}_n) \le C\rho_n\delta_n$ by Cauchy–Schwarz. This proves all assertions and the quantitative transfer bound used below. □

The preceding lemma proves Theorem 1.

### A.2 Covariance and its sampling components

**Lemma A.4** (Covariance identity and variance order)**.** *Under Assumptions 1–5, define*

$$\mathbf{A}_h(t,u) = \mathbf{L}\mathbf{D}_h(t)^{-1}\{\boldsymbol{\theta}_r((u-t)/h) \otimes \mathbf{I}_p\}K_h(u-t).$$

*Then the covariance of* $n^{-1/2}\sum_i \boldsymbol{\Psi}_i$ *is*

$$\begin{aligned}
\mathbf{V}_n(s,t) &= \kappa_{1,n}\mathbf{V}_{h,d}(s,t) + (1-\kappa_{1,n})\mathbf{V}_{h,o}(s,t),\\
\mathbf{V}_{h,d}(s,t) &= \int \mathbf{A}_h(s,u)\mathbf{C}(u,u)\mathbf{A}_h(t,u)^{\mathsf{T}} f(u)\,du,\\
\mathbf{V}_{h,o}(s,t) &= \iint \mathbf{A}_h(s,u)\mathbf{C}(u,v)\mathbf{A}_h(t,v)^{\mathsf{T}} f(u)f(v)\,du\,dv.
\end{aligned}$$

*Let* $\mathbf{e}_0 = (1,0,\ldots,0)^{\mathsf{T}}$ *and define*

$$\begin{aligned}
K_{r,t,h}(z) &= \mathbf{e}_0^{\mathsf{T}}\boldsymbol{\Gamma}_r(t,h)^{-1}\boldsymbol{\theta}_r(z)K(z)\mathbf{1}_{I_{t,h}}(z),\\
\nu_r(t,h) &= \int K_{r,t,h}(z)^2\,dz.
\end{aligned}$$

*There are constants* $0 < c_\nu < C_\nu$ *such that* $c_\nu \le \nu_r(t,h) \le C_\nu$ *uniformly in* $t,h$*. Uniformly on* $\mathcal{T}$*,*

$$\mathbf{V}_n(t,t) = \left\{1 - \kappa_{1,n} + \frac{\kappa_{1,n}\nu_r(t,h)}{hf(t)}\right\}\mathbf{V}_F(t) + O_{\mathrm{op}}(hv_n^2).$$

*In particular, with* $\tau_k^2(t) = [\mathbf{V}_n(t,t)]_{kk}$,

$$cv_n^2 \le \tau_k^2(t) \le Cv_n^2 \qquad (t \in \mathcal{T}, k \le p).$$

*Proof.* Expand the two subject sums. Identical observation indices have score second moment $\mathbf{C}(u,u)$; distinct indices have joint moment $\mathbf{C}(u,v)$ and independent times with density $f(u)f(v)$. There are $m_i$ pairs of the first type and $m_i(m_i-1)$ of the second. Averaging the identity over subjects gives the result. Both covariance components arise from the single functional error process; their distinction concerns observation indices.

Lemma A.1 and the operator inverse identity give, uniformly for $t \in \mathcal{T}$ and $z \in I_{t,h}$,

$$h\mathbf{A}_h(t, t+hz) = \frac{K_{r,t,h}(z)}{f(t)}\mathbf{\Omega}(t)^{-1} + O_{\mathrm{op}}(h).$$

The moment identity

$$\mathbf{\Gamma}_r(t,h)\mathbf{e}_0 = \int_{I_{t,h}} \boldsymbol{\theta}_r(z)K(z)\,dz$$

implies $\int K_{r,t,h} = 1$. The uniform inverse bound for $\mathbf{\Gamma}_r(t,h)$ gives $\nu_r(t,h) \le C$; Cauchy–Schwarz and $|I_{t,h}| \le 2$ give $\nu_r(t,h) \ge 1/2$. Changing variables in the two covariance integrals and using operator-norm Lipschitz continuity gives, uniformly on $\mathcal{T}$,

$$\mathbf{V}_{h,d}(t,t) = \frac{\nu_r(t,h)}{hf(t)}\mathbf{V}_F(t) + O_{\mathrm{op}}(1),$$

$$\mathbf{V}_{h,o}(t,t) = \mathbf{V}_F(t) + O_{\mathrm{op}}(h).$$

The weighted error is at most $C(h+\kappa_{1,n}) = Chv_n^2$. The coordinate variances of $\mathbf{V}_F$ are bounded above by the operator bounds and below by assumption. The scalar coefficient in braces is comparable to $1 + \kappa_{1,n}/h$, uniformly over $0 \le \kappa_{1,n} \le 1$. Absorbing the $O(hv_n^2)$ error proves the variance bounds. □

Write

$$\mathbf{T}(t) = \operatorname{diag}(\tau_1^2(t), \ldots, \tau_p^2(t)), \qquad \mathbf{M}_i(t) = \mathbf{T}(t)^{-1/2}\mathbf{\Psi}_i(t),$$

and $\mathbf{Q}_n(s,t) = n^{-1}\sum_i \mathbb{E}\{\mathbf{M}_i(s)\mathbf{M}_i(t)^\intercal\}$. Thus each coordinate of the matching Gaussian process has variance one, without any requirement that its full covariance be invertible.

### A.3 Oracle moments and empirical covariance

**Lemma A.5** (Subject moments and oracle covariance)**.** *Under Assumptions 1–5,*

$$\max_i \left\| \sup_{t\in\mathcal{T}} \|\mathbf{M}_i(t)\|_\infty \right\|_{L^q} \le \frac{C\sqrt{p}K_n}{hv_n}, \qquad \sup_{t,k} \frac{1}{n}\sum_i \mathbb{E}\{|M_{ik}(t)|^4\} \le Cp^2\Lambda_n.$$

*Moreover,*

$$\mathbb{E}\{R_{Q,n}\} \le Ce_n, \qquad R_{Q,n} = \left\| n^{-1}\sum_i \mathbf{M}_i(s)\mathbf{M}_i(t)^\intercal - \mathbf{Q}_n(s,t) \right\|_{\infty,\mathcal{T}^2}.$$

*For*

$$J_{i,kl}(t) = m_i^{-1}\sum_j K_h(T_{ij}-t)|X_{ijk}X_{ijl}|,$$

*define*

$$\mathcal{J}_n = \max\left\{ \sup_{t,a} \frac{1}{nv_n^2} \sum_i u_{ia}(t)^2, \sup_{t,k,l} \frac{1}{nv_n^2} \sum_i J_{i,kl}(t)^2 \right\}.$$

*There is a fixed $C_J$ for which $\mathbb{P}(\mathcal{J}_n > C_J) \le Ce_n$.*

*Proof.* Every row of $\mathbf{L}\mathbf{D}_h^{-1}$ has uniformly bounded Euclidean norm, so its sum of absolute entries is at most $C\sqrt{p}$. Since every score coordinate is bounded by $CF_i/h$, this proves the envelope bound. For any scalar score entry, expanding its fourth moment into groups with $d \in \{1,2,3,4\}$ distinct observation indices gives

$$\mathbb{E}\{|u_{ia}(t)|^4\} \le C\{1 + (m_i h)^{-3}\}.$$

Indeed, a time appearing $b$ times contributes $\int |K_h(u-t)|^b f(u)\,du \le Ch^{1-b}$, mixed fourth mark moments are uniformly bounded by Hölder's inequality, and the group contains at most $Cm_i^d$ terms. Its contribution is at most $C(m_i h)^{d-4}$. The same argument applies to $J_{i,kl}$. Minkowski's inequality and the $C\sqrt{p}$ row-sum bound therefore give a fourth moment at most $Cp^2\{1 + (m_i h)^{-3}\}$ for each $\Psi_{ik}$. Divide by $\tau_k^4 \asymp v_n^4$ and average.

For covariance products the average coordinate variance is at most $Cp^2\Lambda_n$, by Cauchy–Schwarz and the last bound. Their common envelope is $CpF_i^2/(h^2v_n^2)$, whose maximum over subjects has $L^2$ norm at most

$$\frac{Cp}{h^2v_n^2}\left(\sum_i \mathbb{E}\{F_i^q\}\right)^{2/q} \le \frac{Cpn^{2/q}K_n^2}{h^2v_n^2}.$$

Applying the same expectation maximal inequality to all signed coordinate products and pairs of grid points gives

$$\mathbb{E}R_{Q,n} \le C\left\{p\sqrt{\Lambda_n L_n/n} + \frac{pK_n^2 L_n}{n^{1-2/q}h^2 v_n^2}\right\} \le Ce_n.$$

For completeness, interpolation here can also be controlled uniformly in $p$. The weights satisfy

$$\int \|\mathbf{A}_h(t,u) - \mathbf{A}_h(s,u)\|_{\mathrm{op}}^2 f(u)\,du \le Ch^{-3}|t-s|^2,$$
$$\int \|\mathbf{A}_h(t,u) - \mathbf{A}_h(s,u)\|_{\mathrm{op}} f(u)\,du \le Ch^{-1}|t-s|.$$

These follow from the kernel increments and the $O(h^{-1})$ Lipschitz bound for $\mathbf{D}_h^{-1}$. The covariance identity and the operator bound for $\mathbf{C}$ imply

$$\frac{1}{n}\sum_i \mathbb{E}\{|\Psi_{ik}(t) - \Psi_{ik}(s)|^2\} \le Cv_n^2 h^{-2}|t-s|^2.$$

The reverse triangle inequality in this $L^2$ norm gives $|\tau_k(t) - \tau_k(s)| \le Cv_n h^{-1}|t-s|$. Consequently, samplewise on $\mathcal{T}$,

$$\sup_k |M_{ik}(t) - M_{ik}(s)| \le \frac{C\sqrt{p}F_i}{v_n h^2}|t-s|.$$

The expected covariance-product interpolation error is therefore at most $CpK_n^2\Delta_n/(h^3 v_n^2)$, which is absorbed by $e_n$, since $e_n \ge p^{3/2}\delta_n \ge \sqrt{L_n/n}$.

Finally, each coordinate of $\mathbf{u}_i/v_n$ and $J_{i,kl}/v_n$ has average second moment bounded by a constant. Their fourth moments and product envelopes

obey the same calculation with the row-sum factor omitted (while keeping $L_n$ and $K_n$). Thus the expected maximum fluctuation of their empirical squares is at most

$$C\left\{\sqrt{\Lambda_n L_n/n}+\frac{K_n^2 L_n}{n^{1-2/q}h^2 v_n^2}\right\}\le Ce_n.$$

Markov's inequality at a fixed threshold above the population second-moment bound proves the assertion about $\mathcal{J}_n$. □

### A.4 Fitted residuals and covariance control

Let $\widehat{\varepsilon}_{ij}=Y_{ij}-\mathbf{X}_{ij}^{\intercal}\widehat{\boldsymbol{\beta}}(T_{ij})$, and define $\widehat{\mathbf{u}}_i$ by replacing $\varepsilon_i(T_{ij})$ in $\mathbf{u}_i$ by this fitted residual. Put $\widehat{\boldsymbol{\Psi}}_i=\mathbf{L}\widehat{\mathbf{D}}_h^{-1}\widehat{\mathbf{u}}_i$ and

$$\widehat{\sigma}_k^2(t)=n^{-1}\sum_i \widehat{\Psi}_{ik}(t)^2,\quad \widehat{\mathbf{Q}}_n(s,t)_{kl}=\frac{n^{-1}\sum_i \widehat{\Psi}_{ik}(s)\widehat{\Psi}_{il}(t)}{\widehat{\sigma}_k(s)\widehat{\sigma}_l(t)}.$$

Let $\mathcal{V}_n$ be the event that $\widehat{\mathbf{D}}_h$ is invertible on $\mathcal{T}$ and all fitted coordinate variances are positive on $\mathcal{T}$. On this event define

$$\vartheta_n=\sup_{t,k}\left|\frac{\widehat{\sigma}_k(t)}{\tau_k(t)}-1\right|,\qquad \Delta_{Q,n}=\|\widehat{\mathbf{Q}}_n-\mathbf{Q}_n\|_{\infty,\mathcal{T}^2}.$$

Set both quantities to $+\infty$ on $\mathcal{V}_n^c$.

**Lemma A.6** (Quantitative feasible covariance)**.** *Under Assumptions 1–5, if $e_n\to 0$, then, for all sufficiently large $n$ and $0<u\le 1/2$,*

$$\mathbb{P}\{\vartheta_n+\Delta_{Q,n}>u\}\le Ce_n/u.$$

*In particular,* $\mathbb{P}(\mathcal{V}_n^c) \le Ce_n$ *and the discrepancies are* $O_P(e_n)$*. The fitted-score replacement obeys*

$$\sup_{t,k}\left[\frac{1}{n}\sum_i\left\{\frac{\widehat{\Psi}_{ik}(t)-\Psi_{ik}(t)}{\tau_k(t)}\right\}^2\right]^{1/2} = O_P\{p^{3/2}(\delta_n+h^s)\}.$$

*Proof.* The condition $e_n \to 0$ implies $p\delta_n \to 0$ and hence $\rho_n \to 0$, so Lemma A.3 applies. Define

$$\mathcal{H}_n = \mathcal{E}_n \cap \{R_{Q,n} \le 1,\ \mathcal{J}_n \le C_J\}.$$

Its complement has probability at most $C(\rho_n^2 + e_n)$. Let $b_n^{\mathrm{fit}} = \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|_{2,\mathcal{T}}$ and $J_i^0(t) = \sum_k J_{i,kk}(t)$. Uniform empirical invertibility gives, on $\mathcal{H}_n$,

$$\sup_{t,k}\sum_{a=1}^{P}|[\mathbf{L}\widehat{\mathbf{D}}_h(t)^{-1}]_{ka}| \le \sqrt{P}\sup_t\|\mathbf{L}\widehat{\mathbf{D}}_h(t)^{-1}\|_{\mathrm{op}} \le C\sqrt{p}.$$

The residual identity and weighted Cauchy–Schwarz give, for a score coordinate $a$ corresponding to predictor coordinate $k$,

$$|\widehat{u}_{ia}(t) - u_{ia}(t)| \le b_n^{\mathrm{fit}}\{J_{i,kk}(t)J_i^0(t)\}^{1/2}.$$

Since $J_i^0 = \sum_l J_{i,ll}$, the definition of $\mathcal{J}_n$ implies

$$\sup_t \frac{1}{nv_n^2}\sum_i \|\mathbf{u}_i(t)\|_2^2 \le Cp,$$

$$\sup_{t,a}\frac{1}{nv_n^2}\sum_i|\widehat{u}_{ia}(t)-u_{ia}(t)|^2 \le Cp(b_n^{\mathrm{fit}})^2.$$

Indeed, each summand satisfies

$$n^{-1}\sum_i J_{i,kk}J_{i,ll} \le \{n^{-1}\sum_i J_{i,kk}^2\}^{1/2}\{n^{-1}\sum_i J_{i,ll}^2\}^{1/2} \le Cv_n^2.$$

Use the exact decomposition

$$\widehat{\mathbf{\Psi}}_i - \mathbf{\Psi}_i = \mathbf{L}(\widehat{\mathbf{D}}_h^{-1} - \mathbf{D}_h^{-1})\mathbf{u}_i + \mathbf{L}\widehat{\mathbf{D}}_h^{-1}(\widehat{\mathbf{u}}_i - \mathbf{u}_i).$$

Applying the empirical $L^2$ norm to each coordinate and using the empirical row-sum bound gives a fitted-score discrepancy at most

$$\zeta_n = C\sqrt{p}\,\mathcal{D}_n + Cp^{3/2}(\mathcal{U}_n + h^s),$$

where we used $b_n^{\text{fit}} \le C\sqrt{p}(\mathcal{U}_n + h^s)$. Lemma A.2 and $\rho_n \lesssim p\delta_n$ give

$$\mathbb{E}\zeta_n \le Cp^{3/2}(\delta_n + h^s), \qquad \mathbb{E}\zeta_n^2 \le Cp^3(\delta_n + h^s)^2.$$

No independence of fitted residuals and the fitting observations is used in these inequalities.

Before dividing by fitted variances, write

$$\mathcal{W}_n = \sup_{s,t,k,l} \left| \frac{n^{-1}\sum_i \widehat{\Psi}_{ik}(s)\widehat{\Psi}_{il}(t)}{\tau_k(s)\tau_l(t)} - [\mathbf{Q}_n(s,t)]_{kl} \right|.$$

On $\mathcal{H}_n$, the empirical oracle coordinate second moments are at most 2, because $R_{Q,n} \le 1$. Expanding the difference of products and applying Cauchy–Schwarz therefore gives

$$\mathcal{W}_n \le R_{Q,n} + C\zeta_n + \zeta_n^2,$$

$$\mathbb{E}\{\mathbf{1}_{\mathcal{H}_n}\mathcal{W}_n\} \le C\{e_n + p^3(\delta_n + h^s)^2\}$$

$$\le Ce_n.$$

The integrand is defined to be zero outside $\mathcal{H}_n$. If $\mathcal{W}_n$ is smaller than a fixed constant, its diagonal entries give positive fitted variances and fitted-to-oracle variance ratios in $[1/2, 3/2]$. The square-root and inverse square-root maps are Lipschitz on this interval. Since $|[\mathbf{Q}_n(s,t)]_{kl}| \le 1$,

$$\vartheta_n + \Delta_{Q,n} \le C\mathcal{W}_n$$

on this event. Consequently, for $0 < u \le 1/2$, Markov's inequality and the exceptional-probability bound give

$$\mathbb{P}\{\vartheta_n + \Delta_{Q,n} > u\} \le C(\rho_n^2 + e_n) + Ce_n/u \le Ce_n/u.$$

The last step uses $\rho_n^2 \lesssim p^2\delta_n^2 \le p^3(\delta_n + h^s)^2 \le e_n$ for all sufficiently large $n$. This proves the tail bound, and the fitted-score rate follows from its samplewise bound on $\mathcal{H}_n$ and the vanishing exceptional probability. □

## B. Gaussian and bootstrap approximation

Recall $L_n = \log(np/h)$, $s = r + 1$, and let $\mathcal{F}_n$ denote the sigma-field generated by the observations. All conditional probabilities and expectations below condition on this sigma-field. Put

$$M_{ik}(t) = \Psi_{ik}(t)/\tau_k(t), \qquad Z_{n,k}(t) = n^{-1/2}\sum_i M_{ik}(t),$$

and write

$$Q_n(u,t) = n^{-1}\sum_i \mathbb{E}\{M_i(u)M_i(t)^\top\}.$$

The matching Gaussian process $G_n$, its supremum $S_{G,n}$, the statistics $T_n^\circ, T_n$, and the multiplier process and maximum $G_n^*, S_n^*$ are those defined in Section 3.

The preceding lemmas give the following inputs. With $F_i = 1 + \sup_t \|\mathbf{X}_i(t)\|_\infty^2 + \sup_t \|\mathbf{X}_i(t)\varepsilon_i(t)\|_\infty$, we have $\|F_i\|_q \le CK_n$, where $1 \le K_n \le Cp^{1/q}$. Lemma A.4 gives $\tau_k(t) \asymp v_n$ uniformly. The subject moment bounds imply

$$\left\| \max_k \sup_t |M_{ik}(t)| \right\|_q \le CH_n, \qquad \max_k \sup_t \frac{1}{n}\sum_i \mathbb{E}\{M_{ik}(t)^4\} \le CA_n, \tag{B.1}$$

where

$$H_n = \frac{\sqrt{p}K_n}{hv_n}, \qquad A_n = p^2\Lambda_n, \qquad B_n = \max\{H_n, \sqrt{A_n}\}.$$

The $\sqrt{p}$ row-sum bound for $\mathbf{L}\mathbf{D}_h^{-1}$ gives the score envelope, and Minkowski's inequality gives the fourth-moment factor $p^2$. These estimates use the general dimension bounds under the stated assumptions. Since $h^2v_n^2 \le h^2 + h \le 3/4$, we have $H_n \ge 1$, so a separate constant in the maximum defining $B_n$ is unnecessary.

Use $\gamma_n, e_n$ from (3.2) and (3.5), and write

$$r_{n,1} = \frac{\sqrt{np}\,\rho_n\delta_n}{v_n}, \qquad r_{n,2} = \frac{\sqrt{np}\,h^s}{v_n}.$$

The quantitative version of Lemma A.3, including the event controlling the centered Taylor score, gives, for every $u > 0$,

$$\mathbb{P}\left\{ \left\| \sqrt{n}\ \mathrm{diag}(\tau_k^{-1})(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - Z_n \right\|_{\infty,\mathcal{T}} > Cr_{n,2} + u \right\} \leq C\rho_n^2 + Cr_{n,1}/u. \tag{B.2}$$

In particular, the smoothing term in this bound is deterministic on an event whose complement has probability at most $C\rho_n^2$; an assertion in probability about the smoothing remainder alone would not suffice for the coverage calculation below.

Let $\mathcal{V}_n$ be the event of uniform empirical design invertibility and positive fitted variances. On its complement, extend $T_n, S_n^*, G_n^*$ by zero and use empty bands. The statistic $T_n^\circ$ is extended by zero only when the empirical design fails to be uniformly invertible. On $\mathcal{V}_n$, write

$$\vartheta_n = \max_k \sup_t \left| \frac{\widehat{\sigma}_k(t)}{\tau_k(t)} - 1 \right|, \qquad \Delta_{Q,n} = \|\widehat{Q}_n - Q_n\|_{\infty,\mathcal{T}^2},$$

where $\widehat{Q}_n(s,t) = n^{-1}\sum_i \widehat{M}_i(s)\widehat{M}_i(t)^\intercal$ and $\widehat{M}_{ik} = \widehat{\Psi}_{ik}/\widehat{\sigma}_k$. Set both discrepancies to $+\infty$ on $\mathcal{V}_n^c$. Lemma A.6 supplies the quantitative bound

$$\mathbb{P}(\vartheta_n + \Delta_{Q,n} > u) \leq Ce_n/u, \qquad 0 < u \leq 1/2. \tag{B.3}$$

This is a bound on an unconditional probability over the observations; it is stronger than the corresponding $O_P(e_n)$ assertion.

### B.1 A fine grid and continuous Gaussian processes

Choose a regular grid $\chi_n \subset \mathcal{T}$ containing the endpoints and having mesh at most

$$\Delta_n = n^{-8} h^{10} (pK_n)^{-8}, \qquad \epsilon_n = n^{-2} L_n^{-1/2}.$$

Its cardinality $|\chi_n|$ can be chosen of order $1 + \Delta_n^{-1}$. Since $K_n \le Cp^{1/q}$,

$$\log(2p|\chi_n|) \le CL_n. \tag{B.4}$$

Thus no polynomial lower bound on $h$ is needed. Let $\pi_n(t)$ be the nearest grid point to the left, with the right endpoint mapped to itself, and put $\omega_n(W) = \max_k \sup_t |W_k(t) - W_k(\pi_n(t))|$.

**Lemma B.1** (Oracle increments and discretization)**.** *There is a continuous version of $G_n$. Moreover,*

$$\mathbb{E}\{\omega_n(Z_n)\} \le C\sqrt{np}\, K_n h^{-2} \Delta_n / v_n, \tag{B.5}$$

$$\mathbb{E}\{\omega_n(G_n)\} \le Ch^{-1} \Delta_n \sqrt{L_n}, \tag{B.6}$$

$$\mathbb{P}\{\omega_n(Z_n) > \epsilon_n\} + \mathbb{P}\{\omega_n(G_n) > \epsilon_n\} \le Cn^{-1}. \tag{B.7}$$

*Proof.* The kernel and population inverse satisfy the uniform bounds $\|\mathbf{D}_h(t)^{-1}\|_{\mathrm{op}} \le C$ and $\|\mathbf{D}_h(t)^{-1} - \mathbf{D}_h(s)^{-1}\|_{\mathrm{op}} \le C|t-s|/h$. These follow from population invertibility and the $L^1$ increment bound for the kernel weights, using the operator bounds on the design covariance. The

score envelopes consequently give

$$\max_k |\Psi_{ik}(t)| \le C\sqrt{p}F_i/h, \qquad \max_k |\Psi_{ik}(t) - \Psi_{ik}(s)| \le C\sqrt{p}F_i h^{-2}|t-s|.$$

For completeness, the exact covariance calculation also gives

$$\frac{1}{n}\sum_i \mathbb{E}\{|\Psi_{ik}(t) - \Psi_{ik}(s)|^2\} \le C\{\kappa_{1,n}h^{-3} + h^{-2}\}|t-s|^2 = Cv_n^2 h^{-2}|t-s|^2. \tag{B.8}$$

To verify this, separate equal and distinct observation indices. For $|t-s| \le h$, the squared $L^2$ norm of the difference of the two coefficient-weight kernels is at most $Ch^{-3}|t-s|^2$, and its $L^1$ norm is at most $Ch^{-1}|t-s|$. The diagonal and off-diagonal covariance integrals are therefore bounded by the two terms in (B.8), respectively, using the operator bound on the score covariance. For $|t-s| > h$, the variance bound $\tau_k^2 \le Cv_n^2$ yields the same inequality.

The reverse triangle inequality in the Hilbert space with squared norm $n^{-1}\sum_i \mathbb{E}\{(\cdot)^2\}$ now implies $|\tau_k(t) - \tau_k(s)| \le Cv_n h^{-1}|t-s|$. Together with $\tau_k \ge cv_n$, this gives

$$\max_k |M_{ik}(t) - M_{ik}(s)| \le C\sqrt{p}F_i(v_n h^2)^{-1}|t-s|, \tag{B.9}$$

$$\frac{1}{n}\sum_i \mathbb{E}\{|M_{ik}(t) - M_{ik}(s)|^2\} \le Ch^{-2}|t-s|^2. \tag{B.10}$$

Summing the first inequality and taking expectations proves (B.5). The covariance $Q_n$ is positive semidefinite because it is an average of covariance

kernels. Thus it defines a Gaussian process. Equation (B.10) and the Gaussian fourth-moment identity give a continuous version by the Kolmogorov continuity criterion.

Refine every interval of $\chi_n$ dyadically. At refinement level $j$, there are at most $Cp|\chi_n|2^j$ relevant coordinate increments, each with standard deviation at most $Ch^{-1}\Delta_n 2^{-j}$. The Gaussian maximal inequality and summation over $j \geq 0$ give

$$\mathbb{E}\{\omega_n(G_n)\} \leq Ch^{-1}\Delta_n \sum_{j\geq 0} 2^{-j}\sqrt{\log(2p|\chi_n|) + j} \leq Ch^{-1}\Delta_n\sqrt{L_n}.$$

Markov's inequality proves (B.7). For example, $L_n \leq Cnp/h$, $p, K_n, v_n \geq 1$, and $h \leq 1$ show that each expectation in (B.5)–(B.6), divided by $\epsilon_n$, is at most $Cn^{-1}$. □

The same grid suffices for the expected empirical covariance bounds. Indeed, (B.9) and the subject envelopes bound increments of $M_{ik}(s)M_{il}(t)$ by $CpF_i^2(v_n^2h^3)^{-1}$ times the distance between the two time pairs. Their expected interpolation error is at most $CpK_n^2(v_n^2h^3)^{-1}\Delta_n$ and is absorbed by $e_n$. The logarithm of the number of coordinate-grid pairs is still $O(L_n)$.

**Lemma B.2** (Gaussian supremum bounds)**.** *The distribution function* $F_G(z) = \mathbb{P}(S_{G,n} \leq z)$ *is continuous and*

$$\mathbb{E}\{S_{G,n}\} \leq C\sqrt{L_n}, \qquad \sup_z \mathbb{P}(z < S_{G,n} \leq z + u) \leq Cu\sqrt{L_n} \quad (u > 0).$$

*For* $0 < u \leq 1/2$,

$$\sup_{z \geq 0} |F_G((1 \pm u)z) - F_G(z)| \leq C(uL_n + n^{-1}). \tag{B.11}$$

*Proof.* Every coordinate variance equals one. The Gaussian maximal inequality on the signed grid and Lemma B.1 give

$$\mathbb{E}\{S_{G,n}\} \leq \sqrt{2\log(2p|\chi_n|)} + \mathbb{E}\{\omega_n(G_n)\} \leq C\sqrt{L_n}.$$

Apply the equal-variance anti-concentration inequality in Theorem 3(i) of Chernozhukov, Chetverikov and Kato (2015) to signed coordinates on successively finer nested grids. Their maxima increase almost surely to $S_{G,n}$ and their expectations are bounded by $\mathbb{E}\{S_{G,n}\}$. Taking limits gives the displayed anti-concentration bound and hence continuity of $F_G$. The Gaussian union bound on $\chi_n$, followed by (B.7), gives $\mathbb{P}(S_{G,n} > C_0\sqrt{L_n}) \leq Cn^{-1}$ for a fixed sufficiently large $C_0$. For $z \leq 2C_0\sqrt{L_n}$, anti-concentration bounds a multiplicative shift by $CuL_n$. For larger $z$, both thresholds exceed $C_0\sqrt{L_n}$, giving (B.11). □

## B.2 Proof of Theorem 2

Stack both signs of the values of $M_i$ on $\chi_n$. These independent, centered vectors have dimension $2p|\chi_n|$, average coordinate variances one, average coordinate fourth moments at most $CA_n \leq CB_n^2$, and $q$th moments of their

maximum norm at most $CH_n^q \le CB_n^q$. Theorem 2.5 of Chernozhukov et al. (2022), applied to this signed vector, and (B.4) therefore give

$$\begin{aligned}&\sup_z |\mathbb{P}(\|Z_n\|_{\chi_n} \le z) - \mathbb{P}(\|G_n\|_{\chi_n} \le z)| \\ &\quad \le C\left\{\left(\frac{B_n^2 L_n^5}{n}\right)^{1/4} + \frac{B_n L_n^{3/2-1/q}}{n^{1/2-1/q}}\right\} \le C\gamma_n.\end{aligned}$$

This theorem permits singular full covariance matrices. Its variance condition is a coordinatewise condition, satisfied here by normalization. Lemma B.1 replaces grid maxima by continuous suprema. The resulting shifts are at most $2\epsilon_n$, costing $C\epsilon_n\sqrt{L_n} = Cn^{-2}$ by Lemma B.2. Since $B_n \ge 1$ and $L_n \ge \log 2$, the first term of $\gamma_n$ absorbs $n^{-1}$. Consequently,

$$\sup_z |\mathbb{P}(\|Z_n\|_{\infty,\mathcal{T}} \le z) - F_G(z)| \le C\gamma_n. \tag{B.12}$$

For the estimator, combine (B.2) with (B.12) and Gaussian anti-concentration. For any $u > 0$, the resulting bound is

$$C\{\gamma_n + \rho_n^2 + r_{n,1}/u + (r_{n,2} + u)\sqrt{L_n}\}.$$

Choose $u = (r_{n,1}/\sqrt{L_n})^{1/2}$. The definition of $\gamma_n$ includes the resulting term $(r_{n,1}\sqrt{L_n})^{1/2}$ and the bias term $r_{n,2}\sqrt{L_n}$. Also,

$$r_{n,1}\sqrt{L_n} = \frac{\sqrt{npL_n}\,\rho_n\delta_n}{v_n} \ge \sqrt{p}\,L_n\rho_n,$$

because $\delta_n \geq v_n\sqrt{L_n/n}$. Since $\rho_n \to 0$, $\rho_n^2 \lesssim (r_{n,1}\sqrt{L_n})^{1/2}$ for all sufficiently large $n$. Thus the exceptional-event probability is absorbed, giving

$$\sup_z |\mathbb{P}(T_n^\circ \leq z) - F_G(z)| \leq C\gamma_n. \tag{B.13}$$

### B.3 Conditional bootstrap discretization

**Lemma B.3** (Quantitative conditional discretization)**.** *Let*

$$r_n = \mathbb{P}\{\omega_n(G_n^*) > \epsilon_n \mid \mathcal{F}_n\},$$

*with the zero extension already specified. Under the covariance and linearization rate conditions,*

$$\mathbb{P}(r_n > n^{-1}) \leq C(e_n + n^{-1}), \qquad r_n = O_P(n^{-1}). \tag{B.14}$$

*Proof.* Let $b_n = \sup_{t\in\mathcal{T}} \|\widehat{\boldsymbol{\beta}}(t) - \boldsymbol{\beta}(t)\|_2$ and $\overline{F}_a = n^{-1}\sum_i F_i^a$ for $a = 1, 2$. Use the event $\mathcal{H}_n$ on which the empirical inverse is uniformly bounded, $\vartheta_n \leq 1/2$, and $\sqrt{p}b_n \leq 1$. The quantitative linearization and covariance lemmas give

$$\mathbb{P}(\mathcal{H}_n^c) \leq Ce_n.$$

In particular, the truncated first moment bound $\mathbb{E}[\mathbf{1}_{\mathcal{E}_n} b_n] \leq C\sqrt{p}(\delta_n + h^s)$ on the design event gives $\mathbb{P}(\mathcal{E}_n \cap \{\sqrt{p}b_n > 1\}) \leq Cp(\delta_n + h^s) \leq Ce_n$.

On $\mathcal{H}_n$, the samplewise residual identity gives

$$\|\mathbf{X}_{ij}\widehat{\varepsilon}_{ij}\|_2 \leq \|\mathbf{X}_{ij}\varepsilon_{ij}\|_2 + \|\mathbf{X}_{ij}\|_2^2 b_n \leq C\sqrt{p}F_i.$$

Residuals are fixed as the evaluation point varies. The kernel increment bound and the empirical inverse identity therefore imply

$$\|\widehat{\mathbf{u}}_i(t) - \widehat{\mathbf{u}}_i(s)\|_2 \le C\sqrt{p}F_i h^{-2}|t-s|,$$

$$\|\widehat{\mathbf{D}}_h(t)^{-1} - \widehat{\mathbf{D}}_h(s)^{-1}\|_{\mathrm{op}} \le Cp\overline{F}_1 h^{-2}|t-s|,$$

$$\max_k |\widehat{\Psi}_{ik}(t) - \widehat{\Psi}_{ik}(s)| \le Cp^{3/2}(1+\overline{F}_1)F_i h^{-3}|t-s|.$$

Thus, with $D_n = Cp^{3/2}(1+\overline{F}_1)\overline{F}_2^{1/2}h^{-3}$, the empirical $L^2$ norm of each fitted contribution increment is at most $D_n|t-s|$. The reverse triangle inequality for this empirical norm also gives $|\widehat{\sigma}_k(t) - \widehat{\sigma}_k(s)| \le D_n|t-s|$. Since $\widehat{\sigma}_k \ge cv_n$ on $\mathcal{H}_n$, direct normalization yields

$$\left[\mathbb{E}\{(G^*_{n,k}(t) - G^*_{n,k}(s))^2 \mid \mathcal{F}_n\}\right]^{1/2} \le CD_n|t-s|/v_n.$$

This argument avoids differentiating the estimated variance and retains the same observations used in the residual fit.

Conditionally, the process is Gaussian and is continuous on $\mathcal{V}_n$. The dyadic argument in Lemma B.1 gives, on $\mathcal{H}_n$,

$$\mathbb{E}\{\omega_n(G^*_n) \mid \mathcal{F}_n\} \le CD_n\Delta_n\sqrt{L_n}/v_n.$$

Since $\overline{F}_1 \le \overline{F}_2^{1/2}$, $\mathbb{E}\{(1+\overline{F}_1)\overline{F}_2^{1/2}\} \le CK_n^2$. Conditional Markov's inequality and then unconditional Markov's inequality give

$$\begin{aligned}\mathbb{P}(r_n > n^{-1}) &\le \mathbb{P}(\mathcal{H}_n^c) + n\mathbb{E}(\mathbf{1}_{\mathcal{H}_n} r_n) \\ &\le Ce_n + Cn\epsilon_n^{-1} p^{3/2} K_n^2 h^{-3} \Delta_n \sqrt{L_n}/v_n \\ &\le C(e_n + n^{-1}).\end{aligned}$$

The final inequality follows directly from the chosen mesh and $L_n \le Cnp/h$. The same calculation shows $r_n = O_P(n^{-1})$ because $\mathbb{P}(\mathcal{H}_n^c) \to 0$. □

### B.4 Proof of Theorem 3

On $\mathcal{V}_n$, conditional on the observations,

$$\mathbb{E}\{G_n^*(s)G_n^*(t)^\intercal \mid \mathcal{F}_n\} = \widehat{Q}_n(s,t).$$

In particular, every conditional coordinate variance is one; centering the fitted contributions across subjects is not required for this identity. Proposition 2.1 of Chernozhukov et al. (2022), applied conditionally to the signed grid vectors, gives

$$\sup_z |\mathbb{P}(\|G_n^*\|_{\chi_n} \le z \mid \mathcal{F}_n) - \mathbb{P}(\|G_n\|_{\chi_n} \le z)| \le C(\Delta_{Q,n} L_n^2)^{1/2}.$$

The result permits singular covariance matrices, including the empirical bootstrap covariance. Discretization and Gaussian anti-concentration give

the samplewise bound

$$\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - F_G(z)| \le C\{(\Delta_{Q,n} L_n^2)^{1/2} + n^{-1}\} + r_n \quad \text{on } \mathcal{V}_n. \tag{B.15}$$

Equations (B.3) and (B.14) prove the conditional approximation rate $O_P(\sqrt{e_n}\, L_n)$. Here $e_n \ge p^{3/2}\delta_n \ge \sqrt{L_n/n}$ implies $n^{-1} \lesssim \sqrt{e_n}\, L_n$, so the discretization term is absorbed in the stated rate.

We also need an unconditional approximation for the feasible statistic. On $\{\vartheta_n \le u\}$, where $0 < u \le 1/2$, $T_n^\circ/(1+u) \le T_n \le T_n^\circ/(1-u)$. Using (B.3), (B.13), and (B.11) at deterministic thresholds gives

$$\sup_z |\mathbb{P}(T_n \le z) - F_G(z)| \le C\{\gamma_n + e_n/u + uL_n + n^{-1}\}.$$

Taking $u = (e_n/L_n)^{1/2}$ proves

$$\sup_z |\mathbb{P}(T_n \le z) - F_G(z)| \le C\{\gamma_n + (e_n L_n)^{1/2}\}. \tag{B.16}$$

The triangle inequality now yields

$$\sup_z |\mathbb{P}(T_n \le z) - \mathbb{P}(S_n^* \le z \mid \mathcal{F}_n)| = O_P\{\gamma_n + (e_n L_n^2)^{1/2}\}.$$

All stochastic orders here are over the observations; the conditional distribution is explicitly evaluated given $\mathcal{F}_n$.

## B.5 Proof of Theorem 4

For $0 < u \le 1/2$, put $a_n(u) = C_0\{(uL_n^2)^{1/2} + n^{-1}\}$, with $C_0$ sufficiently large. Equations (B.15), (B.3), and (B.14) imply the deterministic tail bound

$$\mathbb{P}\left\{\sup_z |\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - F_G(z)| > a_n(u)\right\} \le C\{e_n/u + n^{-1}\}. \qquad \text{(B.17)}$$

In particular, this step does not substitute an $O_P$ statement into an unconditional coverage probability.

Let $q_G(v) = \inf\{z : F_G(z) \ge v\}$ and define the conditional bootstrap quantile $\widehat{q}_v$ by the same generalized-inverse convention. On the event in (B.17) where the supremum distance is at most $a = a_n(u)$,

$$q_G(v - a) \le \widehat{q}_v \le q_G(v + a), \qquad a < v < 1 - a. \qquad \text{(B.18)}$$

For the lower inequality, if $z < q_G(v - a)$, then $F_G(z) < v - a$ and the conditional bootstrap distribution is less than $v$ at $z$. For the upper inequality, continuity of $F_G$ gives $F_G\{q_G(v + a)\} = v + a$, so the conditional bootstrap distribution is at least $v$ there. These arguments require no strict monotonicity of either distribution function.

For fixed $\alpha \in (0, 1)$, apply (B.18) at $v = 1 - \alpha$ and apply (B.16) at the two resulting deterministic thresholds. The probability of the exceptional event in (B.17) is added explicitly. This gives

$$|\mathbb{P}(T_n \le \widehat{q}_{1-\alpha}) - (1 - \alpha)| \le C\{\gamma_n + (e_n L_n)^{1/2} + e_n/u + \sqrt{u}L_n + n^{-1}\}.$$

If the total bound is small, take $u = (e_n/L_n)^{2/3}$; otherwise the claimed bound follows by increasing its constant. Since $e_n L_n^2 \to 0$, this choice is eventually at most $1/2$ and $a_n(u) < \min(\alpha, 1-\alpha)$. The last display becomes

$$|\mathbb{P}(T_n \le \widehat{q}_{1-\alpha}) - (1-\alpha)| \le C\{\gamma_n + (e_n L_n^2)^{1/3}\}.$$

Here $(e_n L_n)^{1/2}$ is absorbed by $(e_n L_n^2)^{1/3}$ for all sufficiently large $n$. On $\mathcal{V}_n$, the event on the left is exactly simultaneous coverage over all times and coordinates. The probability of $\mathcal{V}_n^c$ is at most $Ce_n$ by (B.3), and is absorbed by the same bound for any fixed convention for the bands on that event.

## C. Sampling frequency, dimension, and bandwidth

This section records consequences of the main results and sharper inputs when observation counts are comparable. They are supplementary to the linearization, approximation, and coverage argument in the main article.

### C.1 Approximation rates across sampling frequencies

Suppose $c_m m \le m_i \le C_m m$ for fixed positive constants $c_m, C_m$. Writing $m_{H,n} = \kappa_{1,n}^{-1}$, define the effective sample size by

$$N_n = \frac{n}{v_n^2} = \frac{n}{1 + (m_{H,n}h)^{-1}} \asymp n \min(1, mh). \tag{C.1}$$

Thus the scale changes from the number of observations in local windows, $nmh$, to the number of independent subjects, $n$. To preserve this dependence in the approximation bounds, set

$$\delta_n^\dagger = \sqrt{\frac{L_n}{N_n}} + \frac{K_n n^{1/q} L_n^2}{N_n},$$
$$\rho_n^\dagger = p^{q/(2q-2)} \sqrt{\frac{L_n}{N_n}} + \frac{p n^{1/q} L_n^2}{N_n},$$
$$B_n^\dagger = v_n \max\{p, \sqrt{p}\, K_n L_n\},$$
$$e_n^\dagger = \frac{p K_n^2 n^{2/q} L_n^3}{N_n} + p^{3/2}(\delta_n^\dagger + h^s).$$

Let $\gamma_n^\dagger$ be defined by (3.2) with $B_n^\dagger, \delta_n^\dagger, \rho_n^\dagger$ in place of $B_n, \delta_n, \rho_n$.

**Proposition C.1** (Sampling-dependent approximation bounds)**.** *Under Assumptions 1–5 and comparable counts, Theorems 1–4 hold with each rate replaced by its dagger version, under the corresponding rate conditions.*

*Proof.* Let $H_i(t) = m_i^{-1} \sum_j K_h(T_{ij} - t)$. Partition $[0,1]$ into intervals of length between $h/2$ and $h$. Each kernel window intersects at most six intervals. Bernstein's inequality for their binomial counts and a union bound give

$$\left\| \sup_t H_i(t) \right\|_{L^q} \le C \left\{ 1 + \frac{\log(1/h)}{m_i h} \right\} \le C L_n v_n^2.$$

For example, the maximum bin count exceeds $C\{m_i h + \log(1/h) + z\}$ with probability at most $e^{-z}$; integration proves the displayed moment bound. The envelope $F_i$ in Assumption 4 is independent of these counts. Every

design or score entry is bounded by $CF_iH_i(t)$, so its supremum has $L^q$ norm at most $CK_nL_nv_n^2$. The coordinate variance remains at most $Cv_n^2$. The score maximal inequalities in Lemma A.2 therefore give $\delta_n^\dagger$. The spectral design and normalized Euclidean Taylor envelopes are at most $CpL_nv_n^2$ in $L^q$, using independence of $\sup_u \|\mathbf{X}_i(u)\|_2^2$ and the observation counts in the windows. Their matrix variance bounds remain $Cp^{q/(q-1)}v_n^2$. The matrix second-moment inequality thus gives $\rho_n^\dagger$.

The standardized contribution envelope is at most $C\sqrt{p}K_nL_nv_n$. Furthermore,

$$\Lambda_n \lesssim \frac{1+(mh)^{-3}}{\{1+(mh)^{-1}\}^2} \lesssim v_n^2,$$

so its coordinate fourth moments are at most $Cp^2v_n^2$. These give the Gaussian approximation input $B_n^\dagger$ and an empirical covariance bound of order

$$p\sqrt{L_n/N_n} + \frac{pK_n^2n^{2/q}L_n^3}{N_n}.$$

The first term is absorbed by $p^{3/2}\delta_n^\dagger$. The samplewise residual discrepancy bounds are unchanged, giving $e_n^\dagger$. The original increment bounds remain available for interpolation. Substitution into the approximation and quantile proofs establishes the proposition. □

In particular, the leading Gaussian term is

$$\left\{\frac{\max(p^2, pK_n^2L_n^2)L_n^5}{N_n}\right\}^{1/4},$$

and the bias term is $\sqrt{pN_nL_n}\,h^s$. The expressions separate the effects of observation frequency and dimension. Once $mh$ is bounded away from zero, increasing the observation count no longer changes the polynomial order of $N_n$; dimension and smoothing still affect accuracy. This is a statement about the displayed sufficient bounds, not an optimality assertion.

*Proof of Corollary 1.* With $m_i = m$, (C.1) gives $N_n = n/(1 + (mh)^{-1}) \asymp n \wedge (nmh)$ and $m_{H,n} = m$. Also $K_n \le C$, and $h \ge n^{-A}$ implies $L_n \asymp \log n$. The two balances in (3.8) therefore give

$$N_n \gtrsim n^{4/q} L_n^{3-4/q}, \qquad N_n^{3/4} h^s \lesssim L_n^{5/4}. \tag{C.2}$$

In particular $N_n$ grows at least polynomially, and the refined inputs satisfy

$$\delta_n^\dagger + \rho_n^\dagger \le C \left\{ \sqrt{\frac{L_n}{N_n}} + \frac{n^{1/q} L_n^2}{N_n} \right\}, \qquad B_n^\dagger \le C v_n L_n.$$

The first balance implies $\rho_n^\dagger \to 0$. Substitution into the first two terms of (3.2), using $n/v_n^2 = N_n$, gives

$$\left( \frac{(B_n^\dagger)^2 L_n^5}{n} \right)^{1/4} \le C \left( \frac{L_n^7}{N_n} \right)^{1/4},$$
$$\frac{B_n^\dagger L_n^{3/2-1/q}}{n^{1/2-1/q}} \le \frac{C n^{1/q} L_n^{5/2-1/q}}{\sqrt{N_n}}.$$

The stochastic linearization term is at most

$$\left( \frac{\sqrt{npL_n}\,\rho_n^\dagger \delta_n^\dagger}{v_n} \right)^{1/2} \le C (N_n L_n)^{1/4} \left\{ \sqrt{\frac{L_n}{N_n}} + \frac{n^{1/q} L_n^2}{N_n} \right\}$$
$$= C \left\{ \frac{L_n^{3/4}}{N_n^{1/4}} + \frac{n^{1/q} L_n^{9/4}}{N_n^{3/4}} \right\}.$$

Since $q \geq 4$ and $L_n, N_n \geq 1$ eventually, the first term is bounded by $C(L_n^7/N_n)^{1/4}$ and the second by $Cn^{1/q}L_n^{5/2-1/q}/\sqrt{N_n}$. The bias term is $O(\sqrt{N_n L_n}\, h^s)$. Finally,

$$\frac{n^{1/q}L_n^{5/2-1/q}/\sqrt{N_n}}{(L_n^7/N_n)^{1/4}} = \left(\frac{n^{4/q}L_n^{3-4/q}}{N_n}\right)^{1/4},$$

so the first balance in (C.2) absorbs the finite-moment term. The second balance also gives

$$\sqrt{N_n L_n}\, h^s \lesssim \frac{L_n^{7/4}}{N_n^{1/4}}.$$

Thus $\gamma_n^\dagger \lesssim (L_n^7/N_n)^{1/4} \to 0$. Proposition C.1 and the substitutions $N_n \asymp n \wedge (nmh)$, $L_n \asymp \log n$ prove (3.9).

To verify that these same conditions also suffice for band coverage, use the definition of $e_n^\dagger$ and fixed $p, K_n$ to obtain

$$e_n^\dagger L_n^2 \leq C\left\{\frac{L_n^{5/2}}{\sqrt{N_n}} + \frac{n^{2/q}L_n^5}{N_n} + L_n^2 h^s\right\}.$$

Here the term $n^{1/q}L_n^4/N_n$ from $\delta_n^\dagger L_n^2$ is bounded by the displayed second term. By (C.2), the three displayed terms are bounded, respectively, by constant multiples of

$$n^{-2/q}L_n^{1+2/q}, \qquad n^{-2/q}L_n^{2+4/q}, \qquad n^{-3/q}L_n^{1+3/q}.$$

They tend to zero because $L_n \asymp \log n$. The coverage conclusion now follows from Proposition C.1. The width assertion (3.10) follows from the two bounds in Section C.2, using fixed $p$, $\log(p/h) \asymp \log(1/h)$, and $m_{H,n} = m$.

For the stated example, $q = 8$, $r \geq 1$, and $h = n^{-2/5}$ give $n \wedge (nmh) \geq n^{3/5}$ for every $m \geq 1$. This exceeds $n^{1/2}(\log n)^{5/2}$ eventually. Since $s \geq 2$, also

$$\{n \wedge (nmh)\}^{3/4} h^s \leq n^{3/4-4/5} = n^{-1/20},$$

which verifies the second balance. □

## C.2 Covariance transitions and band width

Write $\lambda_n = (m_{H,n}h)^{-1}$. The exact covariance identity in Section A.2 distinguishes identical from distinct observation indices of the same functional error. Its local part has order $\lambda_n$. The distinct-index component is $O(1)$ and has order one as $\kappa_{1,n} \to 0$; it is zero when all $m_i = 1$. Consequently $\lambda_n \to \infty$, $\lambda_n \asymp 1$, and $\lambda_n \to 0$ describe local, mixed, and functional covariance scales. No sampling regime needs to be selected to implement the bootstrap.

For clarity, a Gaussian process with the exact oracle covariance can be constructed as a sum of two independent auxiliary Gaussian processes. Using the weights $\mathbf{A}_h$ from Lemma A.4, these are

$$W_d(t) = \int \mathbf{A}_h(t,u)\mathbf{C}(u,u)^{1/2}\sqrt{f(u)}\, dB(u),$$
$$W_o(t) = \int \mathbf{A}_h(t,u)W(u)f(u)\, du,$$

where $B$ is vector white noise and $W$ is an independent Gaussian process with covariance $\mathbf{C}(u, v)$. Multiplying these processes by $\sqrt{\kappa_{1,n}}$ and $\sqrt{1 - \kappa_{1,n}}$, respectively, gives the covariance of the unstandardized oracle sum. This construction represents a covariance identity; the model still contains only $\varepsilon_i(t)$.

The uniform operator bounds on the covariance and the local weights imply that the standardized full covariance differs from the standardized local one by $O(\lambda_n^{-1})$ when $\lambda_n \to \infty$. When $\lambda_n \to 0$, it differs by $O(h + \lambda_n)$ from the correlation kernel of

$$\mathbf{\Omega}(t)^{-1} W(t), \qquad \mathbb{E}\{W(u)W(t)^{\intercal}\} = \mathbf{C}(u, t).$$

Here $\mathbf{\Omega}(t) = \mathbb{E}\{\mathbf{X}(t)\mathbf{X}(t)^{\intercal}\}$. Applying the same grid comparison as in the bootstrap proof shows that the corresponding Gaussian suprema have distribution errors at most $C\{L_n/\sqrt{\lambda_n} + n^{-1}\}$ and $C\{L_n\sqrt{h + \lambda_n} + n^{-1}\}$, respectively. The first bound requires $\lambda_n^{-1} L_n^2 \to 0$ for convergence and the second requires $(h + \lambda_n) L_n^2 \to 0$. These sufficient conditions make explicit why a pointwise covariance transition alone does not justify a simultaneous distributional limit when dimension grows.

Let $w_{n,k}(t) = \widehat{q}_{1-\alpha}\widehat{\sigma}_k(t)/\sqrt{n}$. Under the conditions of Theorem 4, there are constants $0 < c < C$ such that, with probability tending to one,

$$c\left\{\frac{1}{\sqrt{n}} + \sqrt{\frac{\log(1/h)}{nm_{H,n}h}}\right\} \le \inf_{t,k} w_{n,k}(t),$$

$$\sup_{t,k} w_{n,k}(t) \le C\left\{\sqrt{\frac{\log(2p)}{n}} + \sqrt{\frac{\log(p/h)}{nm_{H,n}h}}\right\}.$$

The same statement holds under the refined rate conditions in Proposition C.1. To see the upper bound, set $Z(t) = \mathbf{\Omega}(t)^{-1}W(t)$ and $\mathbf{B}_h(t,u) = \mathbf{A}_h(t,u)\mathbf{\Omega}(u)f(u)$. The exact reproduction identity

$$\int_{\mathcal{T}} \mathbf{B}_h(t,u)\,du = \mathbf{I}_p$$

follows from the definition of $\mathbf{D}_h(t)$. The weights have uniformly bounded operator-norm $L^1$ norms, and their $L^1$ increments are at most $C|t-s|/h$. The increment covariance of $Z$ has operator norm at most $C|t-s|$. For $|t-s| \le h$, integrate $\{\mathbf{B}_h(t,u)-\mathbf{B}_h(s,u)\}\{Z(u)-Z(s)\}$ over the union of the kernel windows. Its coordinate $L^2$ norm is at most $C|t-s|/\sqrt{h} \le C\sqrt{|t-s|}$. For $|t-s| > h$, use $\max_{k\le p}\|W_{o,k}(t)-Z_k(t)\|_{L^2} \le C\sqrt{h}$. Thus the functional Gaussian component has coordinate increments bounded in mean square by $C|t-s|$ throughout $\mathcal{T}$. The local component has coordinate $L^2$ increments at most $Ch^{-1/2}\min\{1,|t-s|/h\}$. Gaussian entropy bounds over $p$ coordinates give expected suprema at most $C\sqrt{\log(2p)}$ and $Ch^{-1/2}\sqrt{\log(p/h)}$. Gaussian concentration transfers these bounds to fixed quantiles. For the

lower bound, select one coordinate and $\asymp h^{-1}$ points with disjoint kernel windows. The local white-noise component at these points is independent, with variances bounded below by $c/h$. Condition on the independent functional Gaussian component. For each local normal variable, the probability of lying in an interval of a given length is maximized when that interval is centered. Hence, with $J \asymp h^{-1}$ and every $x > 0$, the probability that all selected sums have absolute value at most $x$ is at most $\{2\Phi(x/\sqrt{c\lambda_n}) - 1\}^J$. This gives a fixed-quantile lower bound of order $\sqrt{\lambda_n \log(1/h)}$. A single nondegenerate coordinate gives a lower bound of order $v_n$, which also supplies the stated constant-order term. Uniform variance consistency and the bootstrap quantile bracket transfer both bounds to $w_{n,k}$.

When $p$ is bounded these two orders agree. For growing $p$, the displayed lower bound uses a single coordinate and is deliberately not sharpened in the dimension. The sparse-to-dense variance scale and the cost of simultaneous coverage over many functions are therefore distinct.

### C.3 Bandwidth choice and implementation

For increasing $p$, admissibility is determined by $\gamma_n \to 0$ and $e_n L_n^2 \to 0$, or their refined versions. These conditions include the undersmoothing requirement $\sqrt{npL_n}\, h^s / v_n \to 0$. They are nonempty for growing VCMs:

with $r = 1$, $q = 8$, $m_i = 1$, $h = n^{-1/4}$, and $p \asymp n^a$ for any $0 < a < 1/4$, the general bounds vanish even using $K_n \lesssim p^{1/8}$. Indeed,

$$\gamma_n = O\{\sqrt{p}\, n^{-1/8}\sqrt{\log n}\}, \qquad e_n = O\{p^{3/2} n^{-3/8}\sqrt{\log n}\},$$

which verify both conditions throughout this range. The dimension restriction comes from the displayed sufficient bounds and is not claimed to be sharp.

For the numerical comparisons, whose dimensions are specified in advance, we retain the local linear bandwidth

$$h = \{1 + \log n\}^{-1/2} \max\{n^{-1/4}, (n m_{H,n})^{-1/5}\}. \tag{C.3}$$

For bounded $p$, this rule satisfies (3.7). Indeed $L_n \asymp \log n$. If $n^{-1/4}$ is the larger term, use $v_n \geq 1$ to obtain normalized bias $O\{(\log n)^{-1/2}\}$. If $(n m_{H,n})^{-1/5}$ is larger, use $v_n \geq (m_{H,n} h)^{-1/2}$ to obtain $O\{(\log n)^{-3/4}\}$. This formula specifies the numerical procedure. It is not a universal bandwidth rule for arbitrary growing dimension; the dimension-dependent conditions above must be checked for such sequences. All fits, residuals, and multiplier contributions use the selected common bandwidth. An independent pilot may select among bandwidths for which the displayed errors vanish uniformly: conditioning on that pilot and integrating the coverage bound proves the same conclusion for the inference sample.

For implementation on a grid of mesh $\Delta$, the increment bound used in Section B gives, on the good design and variance event, a conditional expected interpolation error bounded by $C\mathcal{L}_n\Delta\sqrt{L_n + \log(1/\Delta)}$, where

$$\mathbb{E}\{\mathcal{L}_n \, \mathbf{1}_{\text{good}}\} \leq Cp^{3/2}K_n^2/(v_n h^3).$$

It is enough to use the theoretical grid constructed there, whose additional probability error is $O(n^{-1})$. More generally, a grid error term tends to zero whenever

$$\frac{p^{3/2}K_n^2\Delta}{v_n h^3}\sqrt{L_n\{L_n + \log(1/\Delta)\}} \to 0.$$

To verify this assertion, use conditional Markov's inequality twice and Gaussian anti-concentration. Balancing the two Markov errors and the anti-concentration shift yields the cube root of this displayed quantity, which can be added to the coverage bound. This sufficient condition does not certify a fixed grid such as 201 points. The simulations report coverage on their stated grid.

With $B$ independent multiplier draws, the conditional empirical cdf has uniform error at most $\epsilon_B = \sqrt{\log(2/\eta)/(2B)}$ with conditional probability at least $1 - \eta$, by the Dvoretzky–Kiefer–Wolfowitz inequality (Massart, 1990). Adding this cdf error to the same quantile bracket adds at most $C\epsilon_B + \eta$ to the coverage bound, including the multiplier randomness. This separates statistical, grid, and simulation errors.

## D. Proof of the mean-function corollary

We use the assumptions, rates, and mean-model statistics from Section 3.5. All bounds below are uniform in $t, k$; $r, q$ are fixed, and constants are uniform in $n, p, h$ and the counts. The argument specializes the subject-level linearization and covariance bounds before applying any dimension-dependent coordinate-mixing bound.

*Proof of Corollary 2.* Let $H(t) = \mathbb{E}\widehat{H}(t)$ and write the oracle contributions as

$$w_h(t,u) = \mathbf{e}_0^{\mathsf{T}} H(t)^{-1} \boldsymbol{\theta}_r\left(\frac{u-t}{h}\right) K_h(u-t),$$

$$\Psi_{ik}(t) = \frac{1}{m_i}\sum_j w_h(t, T_{ij})\varepsilon_{i,k}(T_{ij}).$$

The population matrix $H$ has fixed size $r+1$ and is uniformly invertible by Lemma A.1, applied to scalar predictor one. The coordinatewise covariance calculation in Lemma A.4 gives $cv_n^2 \le \tau_k^2(t) \le Cv_n^2$. Only the covariance entries involving the specified coordinates occur in these calculations; an operator-norm bound on a $p \times p$ covariance matrix is unnecessary.

Put

$$D_n = \sup_t \|\widehat{H}(t) - H(t)\|_{\mathrm{op}},$$

$$U_n = \sup_{t,k}\left\|\frac{1}{n}\sum_i \frac{1}{m_i}\sum_j K_h(T_{ij}-t)\boldsymbol{\theta}_r\left(\frac{T_{ij}-t}{h}\right)\varepsilon_{i,k}(T_{ij})\right\|_2.$$

For each design entry, mutual independence of the observation times gives variance at most $C\kappa_{1,n}/(nh)$ and maximal centered summand at most $C/(nh)$. Use a design grid of mesh at most $n^{-8}h^{10}$, whose log cardinality is $O\{\log(n/h)\}$, independently of $p$. Each entry of $\widehat{H} - H$ has deterministic Lipschitz constant at most $Ch^{-2}$. Bernstein's inequality, integration of its tail, and interpolation therefore yield

$$\mathbb{E}D_n^2 \leq Ca_n^2.$$

For the scores, the average coordinate second moment is at most $Cv_n^2$, and the common subject envelope has $L^q$ norm at most $CK_n/h$. The maximal inequality and grid interpolation used in Lemma A.2 give

$$\mathbb{E}U_n^2 \leq C\delta_n^2.$$

Moreover, $a_n \leq \delta_n$, since $\log(n/h) \leq L_n$, $K_n \geq 1$, and the finite-moment term in $\delta_n$ dominates $L_n/(nh)$.

On the good-design event $\mathcal{E}_n = \{D_n \leq c_0\}$, inversion is Lipschitz and the Taylor remainder is uniformly bounded by $Ch^s$. Consequently,

$$\|\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|_{\infty,\mathcal{T}} \leq C(U_n + h^s), \qquad \|\mathbf{R}_n\|_{\infty,\mathcal{T}} \leq C(D_nU_n + h^s).$$

Since $\mathbb{P}(\mathcal{E}_n^c) \leq Ca_n^2$ and $\mathbb{E}(D_n U_n) \leq Ca_n\delta_n$, these inequalities prove the linear representation and, for every $u > 0$, the quantitative transfer

$$\mathbb{P}\left\{\sup_{t,k} \frac{\sqrt{n}|R_{nk}(t)|}{\tau_k(t)} > C\frac{\sqrt{n}h^s}{v_n} + u\right\} \leq Ca_n^2 + \frac{C\sqrt{n}\, a_n\delta_n}{v_n u}.$$

The smoothing shift is thus deterministic on an event with a controlled failure probability, as needed for the coverage argument.

Let $M_{ik}(t) = \Psi_{ik}(t)/\tau_k(t)$. The repeated-observation-index moment calculation gives

$$\max_i \left\| \sup_{t,k} |M_{ik}(t)| \right\|_{L^q} \leq \frac{CK_n}{hv_n},$$

$$\sup_{t,k} \frac{1}{n}\sum_i \mathbb{E}|M_{ik}(t)|^4 \leq C\Lambda_n.$$

Because $\kappa_{3,n} \leq \kappa_{1,n}$ and $h \leq 1$, $\Lambda_n \leq C/(h^2 v_n^2) \leq CK_n^2/(h^2 v_n^2)$. Also $h^2 v_n^2 \leq 3/4$ and $K_n \geq 1$. Theorem 2.5 of Chernozhukov et al. (2022) therefore applies to the signed grid with moment scale $K_n/(hv_n)$, giving the first two terms of $\gamma_n^\mu$. The grid in Lemma B.1 has log cardinality $O(L_n)$ because $K_n \leq Cp^{1/q}$. The same fixed-coordinate increment bounds give a continuous Gaussian version, an $O(n^{-1})$ continuum error, and anti-concentration at most $Cu\sqrt{L_n}$. Balancing this last error with the transfer probability gives

$$C\left(\frac{\sqrt{nL_n}\, a_n\delta_n}{v_n}\right)^{1/2} + \frac{C\sqrt{nL_n}\, h^s}{v_n}.$$

The stochastic term is at least a constant times $\sqrt{L_n a_n}$, so it absorbs $a_n^2$ when $a_n \to 0$. The first Gaussian term absorbs $n^{-1}$. This proves (3.13).

For oracle covariance products, the average coordinate variance is at most $C\Lambda_n$. The $L^2$ norm of their envelope maximized over subjects is at most $Cn^{2/q}K_n^2/(h^2 v_n^2)$. Applying the product maximal inequality over coordinate and grid pairs, followed by interpolation, gives expected covariance error at most

$$C\left\{\sqrt{\frac{\Lambda_n L_n}{n}} + \frac{K_n^2 L_n}{n^{1-2/q}h^2 v_n^2}\right\}.$$

The empirical quadratic averages of the oracle scores and local kernel masses are bounded by fixed constants outside an event with probability controlled by this same bound. On this event and $\mathcal{E}_n$, samplewise Cauchy–Schwarz gives

$$\sup_{t,k}\left[\frac{1}{n}\sum_i\left\{\frac{\widehat{\Psi}_{ik}(t)-\Psi_{ik}(t)}{\tau_k(t)}\right\}^2\right]^{1/2} \le C(D_n + U_n + h^s).$$

Here the exact residual identity is $\widehat{\varepsilon}_{ij,k} - \varepsilon_{i,k}(T_{ij}) = -\{\widehat{\mu}_k(T_{ij}) - \mu_k(T_{ij})\}$. The truncated first and second moments of the fitted-score discrepancy are $O(\delta_n + h^s)$ when $e_n^\mu \to 0$, using $a_n \le \delta_n$. All exceptional probabilities are $O(e_n^\mu)$ as well. The covariance-product identity and variance normalization therefore imply

$$\mathbb{P}(\Delta_n^\mu > u) \le \frac{Ce_n^\mu}{u}, \qquad 0 < u \le 1/2, \tag{D.1}$$

where $\Delta_n^\mu$ is the sum of relative variance error and standardized fitted covariance error, set to infinity on invalid samples. This also controls failure of fitted variance positivity.

The conditional discretization proof in Lemma B.3 applies with fixed matrix size $r+1$. On its good event the expected random Lipschitz factor is at most $CK_n^2/(v_n h^3)$, and the event's complement has probability $O(e_n^\mu)$. For the conditional interpolation error probability $r_n$, the same grid and two applications of Markov's inequality give

$$\mathbb{P}(r_n > n^{-1}) \le C(e_n^\mu + n^{-1}), \qquad r_n = O_P(n^{-1}).$$

Conditional Gaussian comparison and (D.1) now prove the first bootstrap assertion. The second follows from the same studentization bracket as in Theorem 3; the unconditional feasible-statistic error is at most $C\{\gamma_n^\mu + (e_n^\mu L_n)^{1/2}\}$. The discretization terms are absorbed because $e_n^\mu \ge \delta_n \ge v_n\sqrt{L_n/n}$.

For coverage, (D.1) and the conditional grid bound imply, for $0 < u \le 1/2$,

$$\mathbb{P}\Big\{\sup_z \big|\mathbb{P}(S_n^* \le z \mid \mathcal{F}_n) - \mathbb{P}(S_{G,n} \le z)\big| > C(\sqrt{u}L_n + n^{-1})\Big\} \le C(e_n^\mu/u + n^{-1}).$$

Bracket the random bootstrap quantile between the two Gaussian quantiles as in (B.18), and add this bad-event probability. The resulting unconditional coverage error is at most

$$C\{\gamma_n^\mu + (e_n^\mu L_n)^{1/2} + e_n^\mu/u + \sqrt{u}L_n + n^{-1}\}.$$

Taking $u = (e_n^\mu/L_n)^{2/3}$ proves (3.14) when the bound is small; otherwise the trivial bound suffices after increasing $C$. The probability of invalid design or variances is $O(e_n^\mu)$ and is included explicitly. Thus the conclusion controls unconditional coverage, including estimation, residual fitting, studentization, and the random critical value. □

### D.1 A high-dimensional example with polynomial tails

Let $V_i$ be a centered Student $t_\nu$ variable, standardized to have variance one, with $\nu > 8$. Independently, let $R_{ik,0}, R_{ik,1}$ be independent Rademacher variables, and set

$$\varepsilon_{i,k}(t) = V_i\{R_{ik,0} + R_{ik,1}\sin(2\pi t)\}.$$

The errors have polynomial tails and $\sup_{t,k}|\varepsilon_{i,k}(t)| \le 2|V_i|$, so their $L^8$ envelope is bounded independently of $p$. Their covariance entries are $C_{kl}(u,t) =$

$\mathbf{1}_{\{k=l\}}\{1+\sin(2\pi u)\sin(2\pi t)\}$, which are uniformly Lipschitz, with coordinate variances at least one. The shared scale $V_i$ permits dependence across coordinates.

Take $q = 8$, local linear fits, $m_i = 1$, and $h = n^{-1/4}$. Then $v_n \asymp n^{1/8}$ and $K_n = O(1)$. The displayed rates give

$$\gamma_n^{\mu} = O\{n^{-3/16}L_n^{5/4} + n^{-1/4}L_n^{11/8} + n^{-1/8}L_n^{1/2}\},$$
$$e_n^{\mu}L_n^2 = O\{n^{-3/8}L_n^{5/2} + n^{-1/2}L_n^{3}\}.$$

If $\log p = o(n^{3/20})$, both vanish, and the first covariance term in $(e_n^{\mu}L_n^2)^{1/3}$ dominates all the displayed contributions to coverage. The resulting error is $O(n^{-1/8}L_n^{5/6})$. The undersmoothing condition is also explicit: $\sqrt{nL_n}\,h^2/v_n \asymp n^{-1/8}\sqrt{L_n} \to 0$. Thus $p = \lceil \exp(n^{\delta}) \rceil$ is admissible for every fixed $0 < \delta < 3/20$ under polynomial moments, even with one observation per subject. This is a sufficient example, not an optimal dimension range.

## E. Reproducibility and additional numerical results

### E.1 Simulation generating mechanism

The factorial design uses $n \in \{200, 400\}$ subjects and $m \in \{5, 20, 80\}$ observations per subject, with 1,000 datasets in each configuration. All observation times are independent Uniform$(0, 1)$. Set $b(t) = (1, \sin(2\pi t), \cos(2\pi t))^{\intercal}$

and define

$$\mathbf{X}_i(t) = \{1, A_{i1}^\intercal b(t)/\sqrt{2}, \ldots, A_{i,p-1}^\intercal b(t)/\sqrt{2}\}^\intercal,$$

where all entries of the $A_{ik}$ are independent standard normals. Independently, generate

$$\varepsilon_i(t) = Z_{i0} + \tfrac{1}{2} Z_{i1} \sin(2\pi t) + \tfrac{1}{2} Z_{i2} \cos(2\pi t).$$

The $Z$ variables are independent draws from one of three laws: $N(0,1)$, Laplace$(0, 1/\sqrt{2})$, or $(\chi^2_5 - 5)/\sqrt{10}$. Here the Laplace parameter is its scale; the chi-square law is centered and standardized by its mean 5 and standard deviation $\sqrt{10}$. All three have mean zero, variance one, and finite moments of every fixed order. Responses are generated as $Y_{ij} = \mathbf{X}_i(T_{ij})^\intercal \boldsymbol{\beta}(T_{ij}) + \varepsilon_i(T_{ij})$, using this single functional error. Thus $\boldsymbol{\Omega}(t) = \mathbf{I}_p$ and $\mathbf{C}(t,t) = 1.25\mathbf{I}_p$ for every law.

For $p = 2$, take $\beta_1(t) = 0.5 + 0.5t$ and $\beta_2(t) = 0.5 + 0.4\sin\{2\pi(t - 0.5)\}$, and target $\beta_2$. For $p = 5$, add $\beta_3(t) = 0.3\cos(2\pi t)$, $\beta_4(t) = 0.5t^2$, and $\beta_5(t) = 0.4e^{t-1}$, and target all five functions jointly. The intercept-only mean comparison uses the same error process and the mean curve specified below. Local linear fits use the Epanechnikov kernel and $B = 1{,}999$ Gaussian multiplier draws. All comparisons evaluate coverage and width on the same 201 equally spaced points in $[0.15, 0.85]$.

## E.2 Computation and comparator formulas

Epanechnikov-weighted local moments are evaluated by exact polynomial prefix sums. Observation times are not binned, and residuals are fitted at every actual observation time, throughout the observation domain. Local Gram eigenvalues and variance positivity are checked; no ridge term or silent regularization is applied. A failed fit is counted as noncoverage. All methods within a comparison use the same dataset and target event.

The simulation bandwidth is (C.3), with multiplicative constant one. It is determined by $n$ and the counts, without using observed coverage or responses. The estimate, residuals, and multiplier covariance use that common value. This is a controlled comparison at specified tuning parameters, not a claim about the relative performance of all possible tuning rules.

For CLZ, put $N = \sum_i m_i$ and use the pooled local linear fit and its residuals. We use the same estimation bandwidth for both methods and for all nuisance kernel estimates; the original CLZ cross-validation choices are not imposed on either estimator. With equal counts and this common bandwidth, the two point estimates agree. Following equations (3.2), (3.5),

and (3.6) of Cao, Liu and Zhou (2018), compute

$$\widehat{f}(t) = N^{-1} \sum_{ij} K_h(T_{ij} - t),$$

$$\widehat{\Omega}(t) = N^{-1} \sum_{ij} K_h(T_{ij} - t) \mathbf{X}_{ij} \mathbf{X}_{ij}^{\intercal},$$

$$\widehat{\Lambda}(t) = N^{-1} \sum_{ij} K_h(T_{ij} - t) \mathbf{X}_{ij} \mathbf{X}_{ij}^{\intercal} \widehat{\varepsilon}_{ij}^2.$$

The pointwise standard error for coefficient $k$ is

$$\left[ \frac{\nu_K}{N h \widehat{f}(t)} \{ \widehat{\Omega}(t)^{-1} \widehat{\Lambda}(t) \widehat{\Omega}(t)^{-1} \}_{kk} \right]^{1/2}, \qquad \nu_K = \int K^2 = 3/5.$$

These formulas reproduce the stated raw kernel-moment normalization for the uniform-design experiments. They are not presented as density-normalized conditional-moment estimates for an arbitrary nonuniform design. The normalization uses the total observation count $N$, not the number of subjects.

Their Section 3 critical-value simulation generates fresh independent $U_1, \ldots, U_N \sim \text{Uniform}(0,1)$ and $g_1, \ldots, g_N \sim N(0,1)$, and records

$$\max_{t \in \mathcal{G}} \frac{\left| \sum_{a=1}^{N} g_a K\{(U_a - t)/h\} \right|}{\sqrt{N h \nu_K}}.$$

We use 20,000 independent draws for each $(n, m)$ combination. The empirical 0.95 quantile gives a single-coefficient band; the 0.99 quantile gives each of the five marginal bands in the Bonferroni comparison. Sorted uniform

times are generated exactly from normalized exponential spacings, and the polynomial kernel sums are evaluated exactly. The calibration maxima are retained. CLZ's sparse-sampling theory does not provide a dense-sampling guarantee.

For the mean comparison, set $X_i(t) \equiv 1$ and

$$\mu(t) = \sin(\pi t) + t + 0.25\{\cos(2\pi t) + \sin(2\pi t)\},$$

a curve also used by Lou and Wu (2026). Retain the same functional error $\varepsilon_i(t)$ generated from Gaussian, Laplace, or adjusted chi-square innovations as in the main experiments. For the Cai–Hu method, let $B_J(t)$ be the cubic B-spline vector with $J$ equally spaced interior knots and repeated endpoint knots. The pooled fit minimizes squared residuals over all $N$ observations. We use the selection rule in equation (5.1) of the public author version, `https://arxiv.org/pdf/2401.17646`, of Cai and Hu (2025):

$$\mathrm{BIC}(J) = \log\left[N^{-1}\sum_{ij}\{Y_{ij} - \widehat{\mu}_J(T_{ij})\}^2\right] + J\log(N)/N,$$

with integer candidates from $\lceil 0.5N^{1/6}\rceil$ to $\lfloor 2N^{1/4}\rfloor$. For the selected $J$, put

$$A = \sum_{ij} B_J(T_{ij})B_J(T_{ij})^{\mathsf{T}}, \qquad s_i = \sum_{j} B_J(T_{ij})\widehat{\varepsilon}_{ij}.$$

The covariance of the spline coefficient estimate is

$$\widehat{V}_\theta = A^{-1}\left(\sum_i s_i s_i^{\mathsf{T}}\right)A^{-1}.$$

The standard error is $\{B_J(t)^\intercal \widehat{V}_\theta B_J(t)\}^{1/2}$. Gaussian coefficient draws from this covariance yield the studentized grid maximum. This comparison uses the same $B = 1{,}999$ and evaluation grid as the proposed method.

Computational checks compare the prefix-sum local fits and subject contributions with direct weighted matrix calculations; verify kernel covariance by quadrature; check both kernel-moment normalizations against direct formulas; verify the finite-observation calibration sums; and check equal-count weighting and the spline subject sandwich. Generator checks verify the single functional-error model and the innovations' centering, variance, and skewness. Further checks verify the functional-error covariance identity by independent quadrature and the rate-consolidation identities algebraically. A fresh rerun of replication zero in all 54 configurations reproduces every inferential result field.

### E.3 Random streams and retained records

There are 54 configurations: three inferential targets, three innovation laws, two subject sample sizes, and three sampling frequencies. Each has 1,000 replications. The configuration's stored `seed_id`, denoted by $j$, and replication $r$ specify the NumPy seed sequence $[20261015, 100000(j+1) + r]$.

Three independent child streams generate data and the two methods' multiplier draws. CLZ calibration uses the independent entropy $[713209, n, m]$. No configuration or replication is omitted according to its outcome.

The supplied `replications.jsonl` records the error-model version, configuration, replication, bandwidth, critical values, coverage indicators, widths, minimum local eigenvalue, and failures. The summary contains Monte Carlo standard errors, Wilson intervals, and paired differences of coverage indicators. The complete experiment contains 54,000 datasets, with both methods fitted to every dataset. The six Gaussian joint cells use seed identifiers 2006, 2007, 2008, 2010, 2011, and 2012 and were recomputed independently with 1,000 datasets each. The current revision retains 36,000 unchanged Gaussian and Laplace records with source provenance and computes all 18,000 adjusted $\chi^2_5$ records afresh. All previous Gaussian joint records are archived; comparisons between independent cohorts and the $m = 20$ versus 80 CP differences, including Monte Carlo uncertainty, are retained with the results. The runner and table generator check the error-model version and reject incomplete or duplicate records. Package versions and execution commands accompany the code.

For each of the six Gaussian joint configurations, an additional check computes weighted least-squares fits and subject contributions directly at

all 201 grid points for all five coefficients, using replication zero. Residual fits are checked at 17 actual observation times, including both extremes. All 1,999 bootstrap maxima agree with the prefix-sum implementation to within $2 \times 10^{-10}$. This check retains the actual counts and bandwidth of each configuration.

### E.4 Additional simulation results

Table 3 reports coverage and width for $\beta_2$ in the two-coefficient model. It complements the joint five-function comparison in main-text Table 1; both tables retain every sampling frequency and error distribution in the factorial design.

Table 4 reports the scalar-mean comparison, in which both procedures retain subject dependence. The growing-dimensional mean result is Corollary 2; its proof is in Section D.

Table 3: Coverage probability (CP) and average width (AW) of nominal 95% bands for $\beta_2$ in the two-coefficient model. Each row uses 1,000 datasets; $m = 5, 20, 80$ denote sparse, mild, and dense sampling.

| | | | Proposed | | CLZ | |
|---|---|---|---|---|---|---|
| Innovations | $n$ | $m$ | CP | AW | CP | AW |
| Gaussian | 200 | 5 | 0.916 | 0.599 | 0.735 | 0.485 |
| | 200 | 20 | 0.925 | 0.460 | 0.326 | 0.244 |
| | 200 | 80 | 0.930 | 0.411 | 0.052 | 0.121 |
| | 400 | 5 | 0.908 | 0.473 | 0.763 | 0.395 |
| | 400 | 20 | 0.925 | 0.347 | 0.389 | 0.197 |
| | 400 | 80 | 0.925 | 0.303 | 0.076 | 0.099 |
| Laplace | 200 | 5 | 0.910 | 0.594 | 0.727 | 0.481 |
| | 200 | 20 | 0.933 | 0.452 | 0.341 | 0.240 |
| | 200 | 80 | 0.936 | 0.409 | 0.053 | 0.121 |
| | 400 | 5 | 0.912 | 0.470 | 0.755 | 0.393 |
| | 400 | 20 | 0.927 | 0.345 | 0.396 | 0.197 |
| | 400 | 80 | 0.929 | 0.302 | 0.083 | 0.099 |
| Adjusted $\chi^2_5$ | 200 | 5 | 0.908 | 0.593 | 0.733 | 0.479 |
| | 200 | 20 | 0.924 | 0.457 | 0.353 | 0.243 |
| | 200 | 80 | 0.925 | 0.411 | 0.059 | 0.121 |
| | 400 | 5 | 0.912 | 0.472 | 0.773 | 0.395 |
| | 400 | 20 | 0.948 | 0.347 | 0.397 | 0.197 |
| | 400 | 80 | 0.922 | 0.302 | 0.083 | 0.099 |

Table 4: Coverage probability (CP) and average width (AW) of nominal 95% bands for the mean function. Each row uses 1,000 datasets; $m = 5, 20, 80$ denote sparse, mild, and dense sampling.

| | | | Proposed | | Cai–Hu | |
|---|---|---|---|---|---|---|
| Innovations | $n$ | $m$ | CP | AW | CP | AW |
| Gaussian | 200 | 5 | 0.937 | 0.609 | 0.939 | 0.534 |
| | 200 | 20 | 0.941 | 0.458 | 0.945 | 0.427 |
| | 200 | 80 | 0.948 | 0.400 | 0.954 | 0.396 |
| | 400 | 5 | 0.942 | 0.479 | 0.938 | 0.379 |
| | 400 | 20 | 0.950 | 0.345 | 0.947 | 0.313 |
| | 400 | 80 | 0.938 | 0.293 | 0.943 | 0.281 |
| Laplace | 200 | 5 | 0.946 | 0.608 | 0.947 | 0.533 |
| | 200 | 20 | 0.941 | 0.457 | 0.952 | 0.427 |
| | 200 | 80 | 0.954 | 0.398 | 0.958 | 0.394 |
| | 400 | 5 | 0.936 | 0.477 | 0.936 | 0.378 |
| | 400 | 20 | 0.939 | 0.344 | 0.938 | 0.313 |
| | 400 | 80 | 0.932 | 0.293 | 0.941 | 0.280 |
| Adjusted $\chi^2_5$ | 200 | 5 | 0.928 | 0.609 | 0.951 | 0.534 |
| | 200 | 20 | 0.925 | 0.455 | 0.937 | 0.425 |
| | 200 | 80 | 0.938 | 0.401 | 0.944 | 0.397 |
| | 400 | 5 | 0.919 | 0.477 | 0.932 | 0.379 |
| | 400 | 20 | 0.945 | 0.344 | 0.947 | 0.312 |
| | 400 | 80 | 0.941 | 0.293 | 0.941 | 0.281 |

### E.5 Multivariate mean experiment

The additional mean experiment uses the fixed factorial design $n \in \{200, 400\}$, $m_i = m \in \{5, 20, 80\}$, and $p \in \{50, 100\}$. With independent Uniform$(0, 1)$ observation times, set

$$\begin{aligned}
\mu_k(t) &= \sin(\pi t) + \tfrac{1}{2}\cos(\theta_k)t + \tfrac{1}{4}\sin(\theta_k)t^2, \\
\theta_k &= 2\pi\{(k-1) \bmod 20\}/20, \\
\varepsilon_{i,k}(t) &= \frac{1}{\sqrt{2}}\big[U_{i0} + Z_{ik0} + \tfrac{1}{2}(U_{i1} + Z_{ik1})\sin(2\pi t) \\
&\qquad + \tfrac{1}{2}(U_{i2} + Z_{ik2})\cos(2\pi t)\big].
\end{aligned}$$

All $U$ and $Z$ variables are mutually independent innovations from the same selected law: $N(0, 1)$, Laplace$(0, 1/\sqrt{2})$, or $(\chi^2_5 - 5)/\sqrt{10}$. The $U$ terms are shared by all coordinates of a subject. Each error coordinate has variance 1.25, and distinct coordinates have contemporaneous correlation 1/2. Thus the Gaussian case is jointly Gaussian, while the other cases retain the selected innovations' heavier tails or skewness. All satisfy the polynomial-moment assumptions of Corollary 2. There is one vector-valued functional error and no added observation noise.

Each $(n, m, \text{innovation law})$ uses 1,000 independent datasets, giving 18,000 independent datasets across the 18 sampling and innovation groups. The two dimensions use nested coordinate prefixes and the same datasets

and multipliers, giving 36,000 dimensionwise evaluations in 36 cells. Every configuration is reported regardless of its coverage. With 1,000 replications, the MCSE of CP near 0.95 is approximately 0.007. Every band uses 999 Gaussian multiplier draws, with one multiplier shared by all coordinates of a subject. Coverage requires every coordinate at all 51 equally spaced evaluation points on $[0, 1]$, including the endpoints. These are grid-coverage measurements. The common bandwidth is (C.3); it does not use responses or coverage outcomes. All observations of subject $i$ evaluate the same error curve $\varepsilon_{i,k}$ defined above.

The implementation shares the scalar local design and fits response coordinates in blocks of at most 128. Exact polynomial prefix sums evaluate residual fits at every actual observation time without forming an observations-by-observations smoothing matrix. Bootstrap maxima are accumulated over all blocks and grid points before taking the empirical 0.95 quantile. No coordinate subsampling, marginal critical values, covariance regularization, or bias correction is used. Independent checks compare the blockwise estimates, residuals, joint bootstrap maxima, critical values, and widths with coordinatewise applications of the scalar code, including endpoints, unequal counts, and shuffled observations. For $n = 400$, $m = 20$, and $p = 50$, complete Gaussian and adjusted chi-square replication-zero

calculations also agree with the original scalar implementation to within $2 \times 10^{-8}$ for all 999 bootstrap maxima.

The seed sequence is $[20261017, n, m, \ell, r]$, where $\ell = 0, 1, 2$ identifies the Gaussian, Laplace, and adjusted chi-square laws, respectively, and $r = 0, \ldots, 999$. Four child streams generate times, common innovations, coordinate-specific innovations, and multipliers. The records in `results/highd-mean/` retain coverage, width, critical values, design and variance diagnostics, conditional smoothing bias divided by the fitted standard error, failures, and time. They also retain code hashes and runtime versions. The fixed design and production-code hashes accompany the results. CP standard errors and Wilson intervals are computed from the 1,000 independent replications in each cell. The new implementation uses Python 3.12.14 and NumPy 2.5.4 with Accelerate BLAS. The mean production run uses six processes alongside four processes for the comparison experiment, with thread-limit environment variables set to one. The retained per-cell time is response-block fitting and calibration time at that dimension; shared design and multiplier setup and initial coefficient generation are excluded. End-to-end times are retained separately. This experiment is independent

of the 54,000-dataset factorial comparison. All multivariate-mean evaluations were recomputed on the 51-point grid; timing records therefore refer to this run. Execution metadata are supplied with the results.

The fixed design contains 18,000 independent subject datasets and 36,000 dimensionwise evaluations. The two dimensions use nested coordinates and the same subject multipliers. The production run computed all 36,000 evaluations in 28.2 minutes using 6 workers.

Replication zero was independently reproduced in all 36 cells. Refining those datasets from 51 to 101 grid points changed 0 coverage judgments; the largest relative critical-value increase was 0.780%. This selected-dataset check does not establish continuum coverage.

### E.6 PBC preprocessing and sensitivity

The raw CSV and its SHA-256 digest are supplied with a provenance record. It contains 1,945 visits from 312 patients. For each analysis horizon, we first restrict the visit days, then remove rows missing any variable needed in that model; bilirubin and alkaline phosphatase must be positive before logarithms are taken. Counts are recomputed after filtering, and weights are $1/m_i$ for those retained counts. Standardization uses observation-level empirical means and standard deviations with divisor $N$. The actual retained

rows, centering and scaling constants, local support counts, estimates, standard errors, and critical values are saved. Thus the exploratory fitted target and its preprocessing are reproducible.

For the 3,000-day horizon, the mean analysis has $N = 1{,}790$ and $m_H = 3.6088$; the six-coefficient analysis has $N = 1{,}741$ and $m_H = 3.4890$. The latter uses a 201-point grid from 450 to 2,550 days. At the primary bandwidth, the nominal positive exclusions for edema are days 733.5–1,311, 1,510.5–1,878, 1,951.5–2,004, and 2,287.5–2,550. For prothrombin time they are days 450–576 and 1,069.5–2,140.5. The albumin exclusions are negative on days 544.5–607.5 and 649.5–775.5. These endpoints refer only to evaluated grid points; they are not estimated change points. The treatment band contains zero throughout, and the log-ALP band is positive throughout.

At bandwidth factor 0.8, the critical value is 3.582 and the albumin band contains zero on the entire grid. At factor 1.2, the critical value is 3.358 and albumin has a negative exclusion extending from day 450 to day 775.5. Treatment and log-ALP retain their respective descriptions across all three factors. The common-random-number seed for the 3,000-day six-coefficient multiplier draws is 530106.

Restricting the horizon to day 2,000 gives 1,518 visits for the mean fit and 1,484 complete visits for the regression, both from 312 patients.

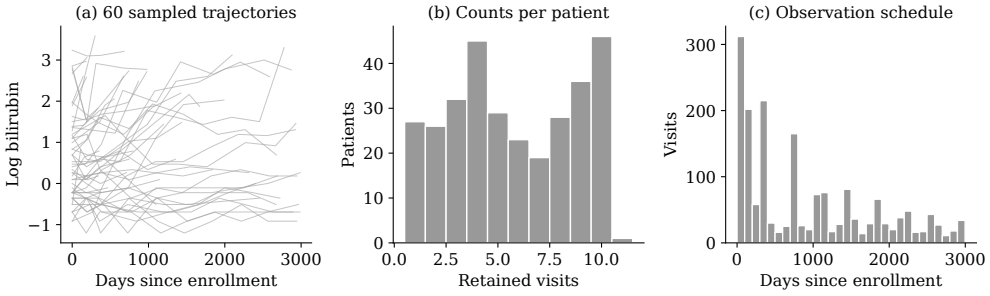



Figure 2: PBC trajectories for 60 patients selected with a fixed seed, retained counts, and visit times up to day 3,000. Baseline and scheduled visits are visible.

After rescaling by 2,000, the regression bandwidth is 0.09598 (about 192 days), $m_H = 3.3330$, and the critical value is 3.576. On the resulting evaluation grid, days 300–1,700, the treatment band again contains zero and the log-ALP band remains positive. The shorter biomarker exclusions vary. Changing the horizon also changes the empirical scaling constants and the composition of observed visits; it is a descriptive sensitivity analysis, not a correction for informative follow-up. None of these comparisons supplies the independence or missingness conditions absent from the clinical observation scheme.

Figure 2 displays the observation schedule and selected trajectories.

For the separate intercept-only analysis in Figure 3, positive observed bilirubin and the 3,000-day restriction leave 1,790 visits from 312 patients.

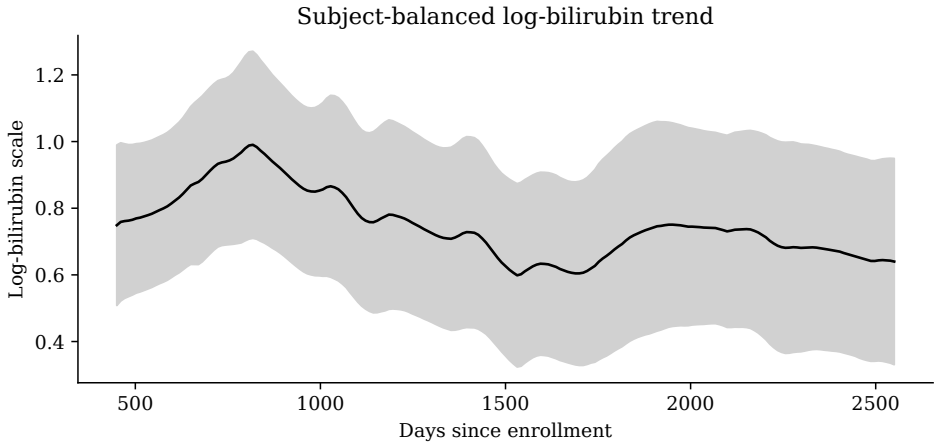



Figure 3: Subject-balanced log-bilirubin trend and nominal 95% band under the working sampling model.

The equal-subject local linear estimate uses $h = 0.09446$, 9,999 multipliers, and the same 201-point evaluation grid as the regression. The critical value is 2.787 and average band width is 0.555. The fit describes the weighted observed visits under the working model. Table 5 summarizes the regression bandwidth sensitivity results.

Table 5: PBC nominal joint-band average widths for three bandwidth factors. The primary factor is $c = 1$.

| Coefficient | $c = 0.8$ | $c = 1$ | $c = 1.2$ |
|---|---|---|---|
| Intercept | 0.774 | 0.647 | 0.602 |
| D-penicillamine | 1.056 | 0.899 | 0.846 |
| Edema | 0.593 | 0.483 | 0.441 |
| Albumin | 0.731 | 0.621 | 0.579 |
| Prothrombin time | 0.767 | 0.670 | 0.629 |
| Log ALP | 0.579 | 0.500 | 0.467 |

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